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WavKAN-CL: An Interpretable and Parameter-Efficient Curriculum Learning Framework **Wavelet**
Kolmogorov–Arnold Networks for Inter-Patient **Arrhythmia Detection** **ECG**
Heartbeat Classification: A Controlled Multi-Seed Evaluation

Venkateramanan Manivannan^a, Ramanathan Lakshmanan^{b,*}

^a*School of Computer Science and Engineering (SCOPE), Vellore Institute of Technology, Vellore, 632014, Tamil Nadu, India*

^b*Department of IoT, School of Computer Science and Engineering, Vellore Institute of Technology, Vellore, 632014, Tamil Nadu, India*

Abstract

Automated arrhythmia classification from ECG signals remains limited by poor model interpretability and reduced inter-patient generalization. In this study, we introduce WavKAN-CL, a framework that combines Wavelet Kolmogorov–Arnold Networks (WavKAN) with a minority-prioritized curriculum learning strategy for arrhythmia classification. In contrast to traditional CNNs, WavKAN replaces fixed activation functions with learnable wavelet basis functions defined on network edges. This enables physiologically grounded modeling of high-frequency morphological features, along with rhythm features that are derived from RR-interval history. To mitigate severe class imbalance in the MIT-BIH Arrhythmia Database, we use a curriculum learning strategy that focuses on minority classes in early training. Under the networks (KANs) with wavelet edge functions are promoted as inspectable models. To our knowledge, they have not been evaluated for ECG heartbeat classification under the inter-patient protocol with repeated runs and matched baselines. We evaluated a 153,045-parameter wavelet KAN (PC-WavKAN) on MIT-BIH under the de Chazal DS1/DS2 protocol the model achieves a mean ventricular (V)recall of 0.88 ± 0.03 across partition against four baselines trained under the same protocol and budget. The design used 20 independent seeds. Despite these gains, minority-class sensitivity remains constrained under strict patient-level separation.

WavKAN-CL contains 95, 189 parameters, which is a >paired seeds, a validation-based architecture selection rule, Holm–Bonferroni correction, effect sizes and confidence intervals. DS2 Macro-F1 was 0.357 ± 0.019 and not statistically distinguishable from any baseline (all Holm-adjusted $p \geq 0.36$; every paired 95% reduction compared to recent deep learning methods such as MAK-Net (6.1M parameters) [1] and DenseNet+GRU architectures [2]. Replacing conventional MLP layers with interpretable KAN layers improves structural transparency while preserving computational efficiency, supporting potential deployment in resource-constrained wearable devices and aligning with Green AI principles [3]. WavKAN-CL is among the 4.1% of recent models that satisfy the strict E3C clinical evaluation criteria [4], achieving this compliance on standard hardware without specialized neuromorphic accelerators% interval within ± 0.03). Supraventricular recall was 0.198 at precision 0.175, and the supraventricular operating point was poor for all five models. In validation ablations of a base configuration, a prior-centred initialisation of the wavelet parameters helped relative to an isotropic one, the mother wavelet did not measurably matter, and self-attention over inter-beat (RR) interval features was harmful. The trained parameters did not support a physiological interpretation: the channel groups are exchangeable, and training moved randomly initialised parameters away from the prior values. Class-balanced sampling raised held-out

*Corresponding author.

Email addresses: venkate.ramanan2024@vitstudent.ac.in (Venkateramanan Manivannan), lramanathan@vit.ac.in (Ramanathan Lakshmanan)

supraventricular recall from 0.137 to 0.222, but not on the validation partition. Re-evaluating all 100 checkpoints without retraining showed two things. An annotation-derived artefact in the RR features of 9.2% of test beats did not inflate performance. Under preprocessing matched to training, the wavelet KAN transferred worse than both convolutional baselines to the St Petersburg INCART and MIT-BIH Supraventricular Arrhythmia databases.

Keywords: Kolmogorov-Arnold Networks Kolmogorov–Arnold networks , Wavelet wavelets , Arrhythmia Classification ECG heartbeat classification , Curriculum Learning inter-patient evaluation , Wearable AI class imbalance , Inter-Patient Generalization reproducibility

1. Introduction

Cardiovascular diseases (CVDs) are currently the leading cause of death worldwide and require reliable and continuous cardiac monitoring. While the Electrocardiogram (The electrocardiogram (ECG) is regarded as the gold standard for the detection of arrhythmias, manual interpretation the reference modality for arrhythmia detection, and manual review of long-term Holter recordings is labor intensive, error-prone and infeasible for large-scale screening. To automate this process, deep learning architectures, in particular Convolutional Neural Networks (CNNs) [5, 2] and Transformers [6] have been widely adopted with high accuracy under controlled experimental conditions.

However, despite these improvements in accuracy, there are three major hurdles that existing state-of-the-art solutions encounter in order to be safely deployed in practice:

Opacity of Black-Box Models: The lack of inherent interpretability in CNNs and Transformers limits the ability of clinicians to associate learned representations with underlying physiological markers, such as P-wave morphology. This lack of interpretability reduces clinical trust and complicates regulatory approval [7]. **Inter-Patient Generalization Gap:** A major limitation of previous studies is recordings does not scale. Convolutional, recurrent and attention-based networks now report high beat-classification accuracy [6, 1]. Much of that literature, however, evaluates under an intra-patient evaluation. This setup involves mixing beats from the same patient in protocol in which beats from one recording appear in both training and test sets and thus leads to inflated reported performance via patient-specific overfitting [8]. Under a strict , which lets a classifier memorise a patient’s morphology and inflates reported accuracy [8, 9, 4]. Under the inter-patient evaluation, models are often shown to have a significant degradation in performance , particularly for minority arrhythmia classes [9, 4]. **Computational Inefficiency:** Architectures such as Vision Transformers (ViT) require millions of parameters, making them unsuitable for battery-powered wearable devices [10]. DS1/DS2 protocol of de Chazal *et al.* [11], reported performance drops sharply, minority-class recall in particular.

To address these interconnected clinical requirements, i.e. interpretability, inter-patient generalization, and edge efficiency, we explore the use of Kolmogorov-Arnold Kolmogorov–Arnold Networks (KANs) [12] as a promising structural foundation. Unlike standard Multi-Layer Perceptrons, KANs use learnable activation functions along edges instead of fixed nodes, making the architecture structurally transparent. However, most existing place learnable univariate functions on network edges. Wav-KAN [13] uses wavelets parameterised by a translation and a dilation as those functions and argues that this makes the learned function space more inspectable. KAN-based ECG models, such classifiers have been reported [1], as have learnable-wavelet convolutional networks for arrhythmia classification [14]. We found no wavelet-KAN evaluated under the de Chazal protocol with repeated seeds, protocol-matched baselines and multiplicity-corrected statistics. We also found no test of whether its parameters admit the physiological reading that the wavelet parameterisation invites.

This paper reports such an evaluation. It is framed as a controlled study, not as MAK-Net [1], rely on B-spline basis functions, which may not optimally capture sharp transient components such as the QRS complex. Furthermore, prior KAN adaptations often lack strict inter-patient validation , efficiency analysis, and robust class imbalance handling.

a method proposal. We ask four questions:

RQ1 Under the inter-patient protocol and a matched training protocol, does a wavelet KAN differ from standard architectures in five-class Macro-F1?

- RQ2** Which design choices measurably matter, as judged on validation data alone: the initialisation of the wavelet parameters, the choice of mother wavelet, the rhythm encoder, an auxiliary morphology branch, and class-balanced sampling?
- RQ3** Do the trained wavelet parameters support a physiological interpretation?
- RQ4** How sensitive are the conclusions to protocol artefacts and to a change of database?

To better capture sharp transient ECG morphology, we substitute B-splines with learnable Mexican Hat Wavelets [15, 13], which inherently localize high-frequency transient ECG features. To address class imbalance, we integrate a minority-prioritized Curriculum Learning strategy [16, 17] that stabilizes decision boundaries early in training.

We therefore propose WavKAN-CL, a wavelet parameterized KAN architecture with minority prioritized curriculum learning for interpretable, generalizable and energy efficient ECG classification. The evaluated model, PC-WavKAN (153,045 parameters), is a wavelet-KAN morphology branch combined with an interval-feature (rhythm) branch. Its one non-standard ingredient is a prior-centred initialisation of the wavelet translations and dilations (PCWI). We propose no new KAN formulation, wavelet parameterisation or training algorithm. The wavelet-KAN layer and its Mexican Hat basis are those of Wav-KAN [13], and interval features and class-balanced sampling are long established. What is new is the evaluation. To our knowledge, this it is the first KAN-based architecture with wavelet edge functions under the strict inter-patient DS1/DS2 protocol.

The contributions of this study include assessment of a wavelet KAN under the de Chazal protocol with repeated seeds, protocol-matched baselines and multiplicity-corrected paired statistics, and the first test of the wavelet-KAN interpretability premise against explicit null models. Our contributions are empirical:

1. **A structurally transparent architecture:** WavKAN replaces opaque layers in MLP architectures with learnable Mexican Hat wavelets. This yields structurally interpretable representations that follow ECG morphology, addressing the black-box opacity barrier without requiring post-hoc attribution mapping.
2. **Robust Inter-Patient Generalization:** We propose a minority-prioritized curriculum learning strategy that mitigates majority-class overfitting and improves robustness under strict DS1/DS2 A 20-seed, seed-paired inter-patient evaluation.
3. **Green AI Efficiency:** Compared to heavily parameterized models like MAK-Net (6.1M parameters) and Vision Transformers, WavKAN-CL operates with only 95,189 parameters—a >comparison against four baselines trained under the same protocol and budget, with a validation-based architecture selection rule (applied after DS2 had been inspected; §4.1), Holm–Bonferroni correction, effect sizes and confidence intervals. It finds no statistically detectable Macro-F1 difference. Every paired 95% reduction in memory footprint—and achieves a median inference latency of 0.78 ms per beat on standard CPU hardware. This offers a rigorously quantified, edge-feasible alternative to support deployment on battery-constrained wearable monitors [3, 10]. confidence interval includes zero and lies within ± 0.03 , and the supraventricular (S) operating point is poor for all five models (§4.2, §4.3).
4. A single-variable component analysis on validation data (§4.1). Initialising the wavelet parameters around fixed prior values (PCWI) improves validation Macro-F1 relative to an isotropic initialisation in the configuration in which the components were ablated. The mother wavelet is not measurably consequential. Self-attention over the five interval features is harmful and was removed.
5. A test of the interpretability premise with explicit null models (§4.7). The channel blocks that receive the three priors are exchangeable, so no block can be assigned a waveform component. Training preserves much of the prior-centred structure, but it moves randomly initialised parameters away from the prior values, so the parameters do not support a physiological reading.
6. Inference-only sensitivity analyses of all 100 checkpoints.

- An annotation-derived artefact in the interval features does not inflate performance (§4.6).
- Under the training preprocessing, with extraction code verified byte-identical on MIT-BIH, the wavelet KAN transfers worse than both convolutional baselines to two external databases (§4.8). An unmatched evaluation of the same checkpoints had suggested the opposite.

A further finding concerns class-balanced sampling. Compared with the inverse-frequency reweighting used by the baselines, it raised DS2 supraventricular recall substantially in two configurations, but this gain did not reproduce on the validation partition for the adopted configuration (§4.4).

2. Related Work

The automated classification of ECG arrhythmias has changed significantly in the past decade. However, the literature shows that there are still consistent methodological differences in terms of evaluation protocols, model transparency, and computational efficiency.

One of the key methodological differences in ECG classification is the choice between

Inter-patient evaluation. Systematic reviews document how prevalent intra-patient and evaluation remains and how strongly it inflates reported performance [9, 4]. Bahrami *et al.* compare inter-patient evaluation. As underscored by systematic reviews by Xiao *et al.* [9] and Silva *et al.* [4], many early deep learning studies adopted the intra-patient paradigm in which heartbeats from a single patient are randomly allocated for training and testing sets. While this approach results in extremely high accuracies ($> 97\%$), it leads to leakage of data, as models tend to memorize the morphological baselines for each individual patient, rather than trying to generalize to physiological markers of arrhythmia [8]. To address this, De Chazal *et al.* [11] proposed the strict inter-patient protocol (intra-patient and patient-specific training of one network [8]. The DS1/DS2 split) for the partition of de Chazal *et al.* [11] is the established remedy for MIT-BIH dataset, where patients in the test set are entirely unseen. Recent state-of-the-art architectures evaluated under this protocol include the convolutional and recurrent neural network (CNN+RNN) by Guo *et al.* [2], the multiscale CNN with attention by Zhou *et al.* [18], and we adopt it unmodified. Architectures evaluated under it include convolutional-recurrent (CNN-RNN) hybrids [2], multiscale attention CNNs [18] and sequence-to-sequence models by Mousavi *et al.* [19]. While these deep models have demonstrated improved inter-patient generalization, they are often based on significantly larger parameter footprints than lightweight edge-deployable models and struggle to maintain high sensitivity on minority classes without extensive synthetic data augmentation (e.g., SMOTE). For example, Mousavi *et al.* [19] report S-class recall above 0.60 under augmented settings, at the expense of models with multi-megabyte memory footprints and synthetic data dependency [8, 19]. [19]. Differences in lead selection, beat extraction, the mapping of annotations to the classes of the Association for the Advancement of Medical Instrumentation (AAMI) and the handling of its rare unclassifiable (Q) class make published accuracies hard to compare directly, so we retrain our baselines under one protocol (§3.5). We give the most direct external comparison, with the protocol’s own source, in §4.3.

The black box nature of standard CNN and Transformer architectures is also a huge challenge to clinical adoption. A recent systematic review by Taleban *et al.* [7] underscored the need to move from post-hoc explanation methods, e.g., Grad-CAM or LIME, which can be computationally expensive and sensitive to noise, towards “intrinsically interpretable” models.

Kolmogorov-Arnold Networks (KANs)[12] have recently become a structurally transparent alternative to Multi-Layer Perceptrons (MLPs), which place learnable univariate activation functions on network edges rather than fixed node activations. Early applications of KANs to ECG signals such as MAK-Net by Zhao *et al.* [1] showed promising results with regard to accuracy using

KANs and learnable wavelet parameterisations. KANs [12] use B-spline bases. However, B-splines may have difficulty separating the fast transient features of physiological signals [13]. To obtain a solution for this, Bozorgasl and Chen [13] proposed Wav-KAN, which makes use of wavelet functions. This is consistent with classical ECG time-frequency analysis theory. In accordance with the theory of Mexican Hat wavelet proposed by Addison [15], the Mexican Hat wavelet has a natural representation of the time-frequency geometry of QRS complex making it a physiologically grounded basis function for interpretable feature extraction. However, existing studies have not systematically investigated wavelet-based KANs with strict inter-patient restriction, parameter efficiency, and class imbalance. edge functions; with parameter counts and FLOPs matched, MLPs generally outperform them outside symbolic formula representation [20]. Wav-KAN [13] substitutes wavelets, including the Mexican Hat, with learnable translation and dilation. The wavelet-KAN layer, the Mexican Hat basis and the argument that a wavelet basis is more inspectable are due to that work and are not claimed here. Learnable wavelet kernels have also been used in convolutional front-ends: WaveletKernelNet [21], for machinery fault diagnosis, learns only their scale and translation, and WaveNet [14] is a SincNet-inspired wavelet CNN for arrhythmia classification on a subject-oriented MIT-BIH benchmark. Unlike those convolutional kernels, a Wav-KAN edge function acts on the *amplitude* of an individual input sample, not on a time window, a distinction that matters for interpretation (§3.3).

Two final, intertwined challenges are severe class imbalance and hardware constraints. Inter-patient datasets show extreme imbalance ratios (e.g. Normal to Fusion beats can be more than 100:1)[4]. Standard cross-entropy training often leads to majority-class collapse. While synthetic augmentation is a common technique, an alternative is Curriculum Learning [16], which takes a dynamic approach in which the sampling distribution is modified during training. As originally formalized by Bengio et al. [16], curriculum strategies are a form of continuation method for non-convex training criteria, MAK-Net [1] combines spline KAN layers with multiscale convolution and a bidirectional gated recurrent unit (BiGRU) for imbalanced ECG classification. It evaluates under an intra-patient protocol. Initialisation of KANs is an active question: Rigas *et al.* [22] compare variance-preserving and guide the optimization process towards better local minima as an implicit regularizer that reduces the generalization error.

This is particularly relevant in our setting, where power-law schemes for spline-KAN coefficients. PCWI addresses a different quantity, the decision boundaries of minority classes are not well-initialized when using normal cross entropy training. Notably, Bae & Shin [23] provided empirical evidence that standard contrastive learning trained directly in unbalanced medical data actually degraded performance (accuracy 74.66% vs. 80.93% baseline), but that their Improved Contrastive Learning (ICL) approach, incorporating curriculum phasing and class distribution-aware proxy sampling, restored and exceeded baseline performance (85.01%). Similarly, Schmale et al. [17] demonstrated that curriculum strategies that prioritized rare and complex medical phenotypes can stabilize the decision boundaries without the risk of producing physiologically invalid synthetic beats. These findings provide motivation for our own wavelet translation and dilation, and leaves coefficient initialisation at a standard default.

Class imbalance. In a systematic study on image benchmarks, Buda *et al.* [24] found oversampling to be the dominant remedy for class imbalance in convolutional networks. They compared it with undersampling, two-phase minority-first curriculum strategy.

training and threshold adjustment. Kang *et al.* [25] compare instance-balanced, class-balanced and progressively balanced sampling for long-tailed recognition. Curriculum approaches vary the sampling distribution or loss weighting over training [16, 23, 17], and adaptive logit adjustment is a logit-level alternative studied for long-tailed multi-label ECG diagnosis [26]. Our training script implements a two-phase schedule of this kind. The selected checkpoint always came from its first, class-balanced phase (§3.4), so what our comparison measures is class-balanced sampling against inverse-frequency reweighting.

Furthermore, deploying these models in wearable Holter monitors requires strict adherence to “Green AI” principles. Farag [3] and Elsheikhy et al. [10] have emphasized the need for reducing memory footprints for

Efficient inference. Small memory footprints are argued for on-device edge inference. Our proposed WavKAN-CL addresses these requirements together in a unified architectural and training framework, combining the structural

transparency of wavelets, stability of curriculum learning and the parameter efficiency required for edge deployment. ECG inference [3, 10]. We report measured size and x86 CPU latency, and say which hardware class was measured (§4.9).

3. Data and Methods

The ECG signal is segmented into beat-centered windows $x_i \in \mathbb{R}^T$ (where $T = 360$ samples, representing 1000 ms at a sampling rate of

3.1. Data and Preprocessing

Each heartbeat appears in the ECG as a P wave (atrial depolarisation), a QRS complex (ventricular depolarisation, whose tallest peak is the R-peak) and a T wave (ventricular repolarisation); the RR interval is the time between consecutive R-peaks. We use the 44 non-paced records of the MIT-BIH Arrhythmia Database [27, 28, 29]. The paced records 102, 104, 107 and 217 are excluded under the protocol. Channel 0 is used. It is the modified lead II (MLII), a Holter adaptation of limb lead II, in every record except 114, where channel 0 is the chest lead V5. Each record is cleaned over its full length with the neurokit method of NeuroKit2: a 0.5 Hz fifth-order Butterworth high-pass filter followed by a 7-sample moving average, both applied forward and backward (zero phase). The library sizes the moving average for 50 Hz mains; at 360 Hz) with RR-interval histories $r_i \in \mathbb{R}^K$ ($K = 5$) in order to learn a mapping function $f : (x_i, r_i) \rightarrow y_i$ Hz it also acts as a low-pass filter, with -3 dB near 16.5 Hz and 34 dB attenuation at 60 Hz. A 360-sample window (1 s at 360 Hz) is then extracted at each annotated beat, except 57 beats that lie too close to a record boundary. The window spans 90 samples (250 ms) before and 270 samples (750 ms) after the annotated R-peak, so the R-peak sits at sample 90. Each window is z -scored with its own mean and standard deviation; a guard against degenerate variance discarded no window. No normalisation statistic is estimated across beats, records or splits.

Interval features. Each beat i also receives a five-element vector

$$r_i = [RR_{-2}, RR_{-1}, RR_0, RR_{+1}, RR_{+2}], \quad (1)$$

where $y_i \in \{N, S, V, F, Q\}$ in a supervised multi-class classification problem. The objective is to maximize the inter-patient generalization under conditions of severe class imbalance ($N \gg V > S > F$).

RR_0 is the interval ending at beat i , RR_{-1} and RR_{-2} are the two intervals before it, and RR_{+1} and RR_{+2} are the two intervals after it. Intervals are in seconds, clipped to $[0.2, 3.0]$ s, and set to 0.8 s where they would extend before the first or after the last annotation of a record. They are computed between consecutive entries of the database annotation file. Those entries include non-beat annotations such as rhythm-change, noise, artefact, comment and non-conducted P-wave markers, so a small fraction of intervals are annotation-to-beat gaps rather than beat-to-beat intervals. §4.6 quantifies this artefact and its effect on every reported model. It is present identically in every model’s input, because every model receives the same features.

The proposed system

Temporal scope. Three properties make the task *offline* beat classification rather than streaming classification. The interval vector contains two intervals that follow beat i . The window’s 750 ms post-R segment contains the next annotated beat for 48.8% of DS2 beats. The filter is zero-phase and applied per record. We therefore do not describe the model as real-time or streaming. A deployment would need a lookahead buffer and a causal filter.

3.2. Inter-Patient Split

We use the DS1/DS2 partition of de Chazal *et al.* [11] and split DS1 by record into training and validation partitions. DS1 training uses the 18 records 101, 106, 108, 109, 112, 114, 115, 116, 118, 119, 122, 124, 201, 203, 205, 207, 215 and 220. DS1 validation uses the 4 records 208, 209, 223 and 230, which carry every model and architecture selection decision. DS2, held out, comprises the 22 records 100, 103, 105, 111, 113, 117, 121, 123, 200, 202, 210, 212, 213, 214, 219, 221, 222, 228, 231, 232, 233 and 234. The partition is by record, and record disjointness is asserted programmatically and covered by a regression test. One subject contributes to two partitions. Records 201 (training) and 202 (DS2) come from the same subject, a known property of the de Chazal partition that we retain unmodified for comparability with prior work, so DS2 is patient-disjoint from training for 21 of its 22 records. §4.2 reports the primary comparison with record 202 removed.

Annotations are mapped to the five classes of the AAMI EC57 standard [11, 30] (Table 1): N (normal, bundle-branch-block and escape beats), S (supraventricular ectopic beats), V (ventricular ectopic beats), F (fusion of a ventricular ectopic and a normal beat) and Q (unclassifiable beats; the paced-beat symbols that AAMI also maps to Q do not occur in these records). The Q class has only 15 beats in these records (6 in training, 2 in validation, 7 in DS2), so no per-class conclusion can be drawn for it. The four validation records hold 386 of DS1’s 414 fusion beats, 372 of them in record 208, which leaves 28 F beats for training. Fig. 1 summarises how each partition and each external database is used.

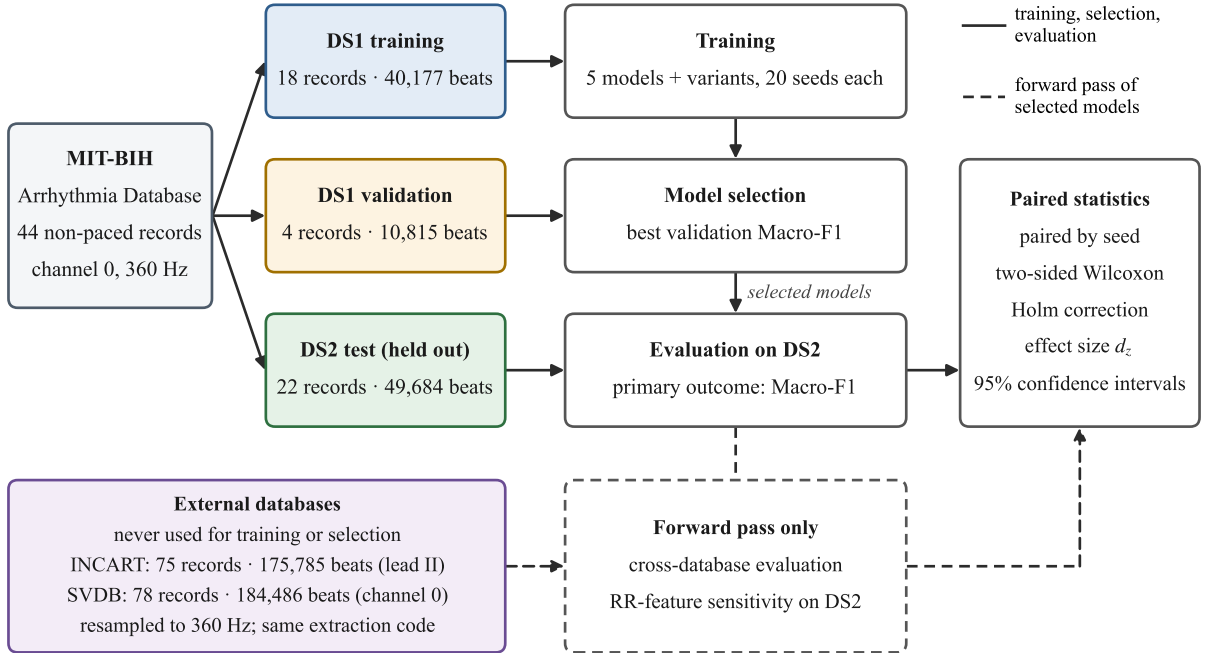


Figure 1: Evaluation protocol. Training records fit the models, the four validation records determine every checkpoint and configuration choice, and DS2 reports the selected models (DS2 metrics of non-adopted variants were also computed during development; §4.1). The external databases and the RR-feature sensitivity analysis use forward passes of the selected models only. Counts are read from the split definition and the extraction outputs.

3.3. PC-WavKAN

Table 1: Class distribution and AAMI mapping.

Class	Included Annotations	Train	Val	Test (DS2)	Total	Ratio (vs N)
N	Normal, L, R, e, j	37,337	8,502	44,232	90,071	1.0 : 1
S	A, a, J, S	485	458	1,837	2,780	32.4 : 1
V	V, E	2,321	1,467	3,220	7,008	12.9 : 1
F	F	28	386	388	802	112.3 : 1
Q	/, f, Q	6	2	7	15	6,004.7 : 1
Total		40,177	10,815	49,684	100,676	

The model (Fig.2) is different from the conventional CNN-based architectures as it employs a dual-branch feature fusion architecture combining a learnable wavelet backbone with a rhythm-aware MLP 2) has a morphology branch and a rhythm branch whose outputs are concatenated and classified.

Morphology branch

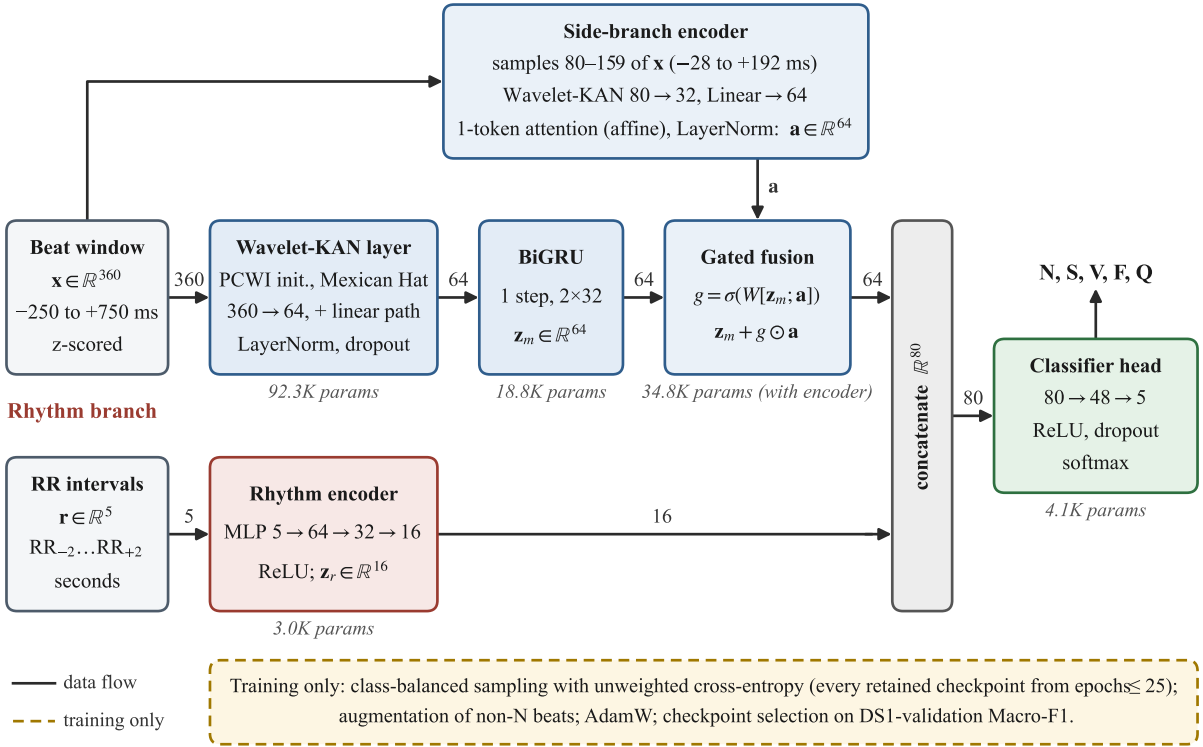


Figure 2: **Hybrid WavKAN Architecture.** Dual-branch architecture that combines the morphological features of WavKAN backbone PC-WavKAN as implemented, with tensor dimensions and the rhythm features of RR encoder per-block trainable parameters (153,045 in total). The use of Morphology branch: the curriculum scheduler beat window passes through a wavelet-KAN layer with prior-centred initialisation (training only PCWI) gives precedence and a linear residual path, followed by layer normalisation, dropout and a bidirectional GRU applied to a single time step. A side branch encodes raw-beat samples 80–159 (28 ms before to 192 ms after the exposure R-peak) into a with a wavelet-KAN layer, a projection and a single-token attention block; $\mathbf{a}$ is added to minority classes the GRU output through a learned sigmoid gate g . Rhythm branch: an MLP encodes the five RR intervals. The two branch outputs are concatenated and classified by an $80 \rightarrow 48 \rightarrow 5$ head. The mechanisms in early epochsthe dashed band act during training only.

The morphological features $\mathbf{z}_m$ are extracted with the help of a dense Wavelet Kolmogorov-Arnold Network (WavKAN) layer. Unlike conventional networks that learn scalar weights, WavKAN learns parameters of a wavelet basis function ψ on each edge. The activation function takes the form of the Mexican Hat wavelet function $\psi(t)$, which is defined as the negative

normalized second derivative of

Wavelet-KAN layer. Following Wav-KAN [13], every edge (j, k) carries its own Mexican Hat function $\psi(u) = (1 - u^2)e^{-u^2/2}$. The layer maps $x_i \in \mathbb{R}^{360}$ to $\mathbb{R}^{64}$:

$$z^{(k)} = \sum_{j=1}^{360} w_{j,k} \psi\left(\frac{x_j - \mu_{j,k}}{|\gamma_{j,k}|}\right) + \lambda \sum_{j=1}^{360} v_{j,k} x_j, \quad k = 1, \dots, 64, \quad (2)$$

with learnable translation $\mu_{j,k}$, dilation $\gamma_{j,k}$ (the implementation adds 10^{-6} to $|\gamma_{j,k}|$) and weight $w_{j,k}$, and a linear residual path with a fixed gate $\lambda = 0.1$, the implementation's default, which was not tuned in this study. ψ is evaluated on the *amplitude* x_j of input sample j , so μ and γ are an amplitude offset and an amplitude width. They are not a temporal position or duration. The output is layer-normalised, passed through dropout ($p = 0.2$), and fed to a bidirectional GRU with 32 units per direction. The 64 activations form a single token, so the GRU executes one step in each direction from a zero state. It therefore acts as a Gaussian function. The choice of the Mexican Hat is motivated to a certain extent theoretically: as opposed to B-splines, which are smooth local polynomials, and have problems with abrupt transitions, gated transformation of the channel vector and integrates no temporal context. It has 18,816 parameters (12.3% of the model), of which the Mexican Hat behaves as a high-frequency transient detector. It more closely matches the sharp transient morphology of the QRS complex than other approaches such as Haar (being too discontinuous) or Morlet (encoding oscillating properties which are inappropriate for a single spike of heartbeats). Furthermore, Addison [15] showed that all derivatives of Gaussian wavelets give rise to modulus maxima lines that are guaranteed to be continuous across scales at signal singularities, which makes them ideal for isolating sharp QRS boundaries in the time-frequency domain.

The WavKAN layer maps the beat vector $x_i \in \mathbb{R}^{360}$ to a compact representation $z_{kan} \in \mathbb{R}^{64}$ through a fully-connected wavelet transform, where each of the 360×64 edges applies an independently parameterized wavelet function. 6,144 recurrent weights do not affect the output, because they multiply the zero initial state.

Each of the

Prior-centred wavelet initialisation (PCWI). The 64 output channels are trained to learn independent values of translation (μ) and dilation (γ) parameters in an end-to-end manner:

$$z_{kan}^{(k)} = \sum_{j=1}^{360} w_{j,k} \cdot \psi\left(\frac{x_j - \mu_{j,k}}{\gamma_{j,k}}\right) \quad \text{for } k = 1 \dots 64$$

split into three blocks, and every edge feeding block c is initialised around a fixed pair $(\bar{\mu}_c, \bar{\gamma}_c)$:

$$\mu_{j,k} \sim \mathcal{U}(\bar{\mu}_c - \epsilon_c, \bar{\mu}_c + \epsilon_c), \quad \gamma_{j,k} \sim \mathcal{U}(\bar{\gamma}_c - \delta_c, \bar{\gamma}_c + \delta_c). \quad (3)$$

The WavKAN output is normalized using LayerNorm and passed through a Bidirectional GRU (Channels 1–32 use $(0.0, 0.05)$ with $(\epsilon, \delta) = (0.02, 0.005)$, channels 33–48 use $(-0.10, 0.12)$, and channels 49–64 use $(+0.10, 0.10)$, the last two with $(\epsilon, \delta) = (0.03, 0.01)$). The values were chosen with reference to the relative amplitude and sharpness of the QRS complex and the P and T waves on the z -scored scale, and we refer to the blocks by those names (QRS-, P- and T-prior blocks). Without PCWI, all edges use the isotropic initialisation $\mu \sim \mathcal{U}(-0.3, 0.3)$, $\gamma \sim \mathcal{U}(0.05, 0.20)$. The weights $w_{j,k}$ and $v_{j,k}$ use Kaiming-uniform initialisation in both cases.

Two properties limit what PCWI can encode. First, each block's prior is shared by all 360 input samples, so no prior is tied to the temporal location of a waveform component. Second, the 64 channels are exchangeable. They feed a layer normalisation initialised identically across channels and a recurrent layer whose input weights are drawn independently and identically across channels. Permuting the channels together with the corresponding downstream weights leaves the network function unchanged, which we verified numerically (maximum output difference 10^{-7}). A network in which the P- and T-block priors are exchanged is therefore an exact relabelling of one in which they are not, with the same initial distribution. What PCWI

determines is the set of initial (μ, γ) values, 32 hidden units per direction)[1, 19] to produce the morphological embedding $z_m \in \mathbb{R}^{64}$. The GRU is used as a nonlinear projection, which takes the wavelet features as conditions before fusing them with the rhythm branch. channels narrow and centred at zero and 16 each slightly offset and wider, not an assignment of channels to waveform components. We name it *prior-centred* for that reason.

Micro-Architecture of WavKAN. In contrast to the conventional MLP where the activation of the nodes is fixed, in WavKAN, learnable wavelet functions $\psi(t; \mu, \gamma)$ are placed on edges. Each of the edges learns its own translation (μ) and dilation (γ) parameters enabling extraction of multi-scale morphological features corresponding to QRS geometry.

Rhythm information is important for the differentiation of S-class arrhythmia. However, raw RR intervals differ greatly among patients due to inter-patient physiological variability. To mitigate this inter-patient variability, we use a Dynamic Normalization strategy adapted from Farag [3]:

1. **Input Vector (R_{seq}):** A sequence of 5 consecutive RR intervals calculated with annotated R-peaks:

$$R_{seq} = [RR_{i-2}, RR_{i-1}, RR_i, RR_{i+1}, RR_{i+2}].$$

2. **Normalization:** Each of the sequences is normalized by dividing each interval by the local moving average (RR_{local}) of the preceding 10 beats:

$$r_i = \frac{R_{seq}}{RR_{local}}$$

The normalized 5-dimensional vector is passed through a three-layered MLP ($5 \rightarrow 64 \rightarrow 32 \rightarrow 16$) to produce the rhythm embedding $z_r \in \mathbb{R}^{16}$.

The concatenated feature vector $z = [z_m; z_r] \in \mathbb{R}^{80}$ is now passed on to a final classifier ($80 \rightarrow 48 \rightarrow 5$). The total number of trainable parameters is 95

Gated side branch. A side branch reads raw-beat samples 80–159, which with the R-peak at sample 90 spans 28 ms before to 192 ms after the R-peak, i.e. the QRS complex and early ST segment. The branch was designed to cover the P-wave, which precedes the QRS complex, but its sample range was fixed for an R-peak at sample 180, and it does not reach the P-wave in the window as extracted. We therefore describe it by what it reads. The samples pass through a randomly initialised wavelet-KAN layer ($80 \rightarrow 32$), are projected to 64 dimensions, and pass through a four-head attention block over a single key/value token. With one token the attention softmax is degenerate, so at inference the block is a learned affine map of that token that does not depend on the query; its 8,189. This translates to a reduction of more than 95% compared to recent KAN and deep learning baselines such as MAK-Net (6.1M parameters)[1] and DenseNet+GRU architectures [2], supporting deployment on resource-constrained edge devices, consistent with Green AI principles [10]. 320 query and key parameters do not affect the output. A sigmoid gate then combines the result with the GRU output. The branch has 34,816 parameters.

Standard curriculum learning progresses from easy to hard samples. Schmale et al. [17] applied this paradigm to the classification of ECG traces by training first on (easy) synthetic signals and then (hard) real signals and showed modest, but consistent improvements in accuracy. However, their approach addresses data source difficulty rather than class imbalance. In severely imbalanced settings, majority Normal (N) beats dominate early gradient updates regardless of source, leading to majority bias. We therefore propose a fundamentally different curriculum axis: a two-phase Minority-First Discovery Schedule that structures training by class rarity rather than source complexity:

1. **Phase 1 (Discovery):** During the first 30% epochs of training, the model is trained on the minority classes only (S, V, F, Q) with no training data of Normal (N) beats. This forces the network to learn discriminative features of the rare phenotypes of arrhythmia before being exposed to the predominant majority class. Standard cross-entropy loss is used in this phase.
2. **Phase 2 (Consolidation):** For the remaining 70% of epochs, training is performed on the whole data set using cross-entropy loss with class weights (minority classes weighted proportional to the reciprocal of their frequency with additional weight on the S-class (a $10\times$ weight multiplier) to mitigate its morphological similarity to Normal beats).

This schedule is only used during the training phase, the inference is done under the natural distribution of the class.

To ensure realistic evaluation and prevent the common “intra-patient” bias observed in many deep learning studies [8, 9], we follow the inter-patient paradigm as outlined by De Chazal et al. [11]. The MIT-BIH Arrhythmia Database is split into two independent sets, DS1 and DS2 as described in Table ?? . Crucially, records 201

Rhythm branch and classifier. The interval vector is encoded to 16 dimensions by a multilayer perceptron (MLP; $5 \rightarrow 64 \rightarrow 32 \rightarrow 16$, ReLU). An alternative encoder that projects each interval to a 16-dimensional token and applies two-head self-attention was also evaluated (§4.1). The 80-dimensional concatenation is classified by an $80 \rightarrow 48 \rightarrow 5$ head. With the MLP rhythm encoder the model has 153,045 trainable parameters, and 202, originating from the same subject, are intentionally separated across DS1 154,325 with the self-attention encoder. Results outside §4.1 use the 153,045-parameter configuration, except three that use the 154,325-parameter base configuration and say so: the pre-specified sampling test (§4.4), the interval ablation (§4.6) and DS2 to prevent data leakage [9]. To enable hyperparameter tuning and early stopping without accessing test data, DS1 was further divided into training and validation subsets following Takalo-Mattila et al. [5]. All reported test results are computed exclusively on DS2, which ensures evaluation on strictly unseen patients. the trained isotropic null (§4.7).

Inter-Patient Data Splitting Protocol (DS1/DS2). The partition separates patients into two groups. Records 201 and 202 (from the same subject) are intentionally separated across DS1 and DS2 in order to impose a strict inter-patient evaluation [3].

Raw annotations from the MIT-BIH database were mapped to the five AAMI EC57 super-classes: Normal (N), Supraventricular (S), Ventricular (V)

3.4. Training

Table 2 collects the settings. Training uses AdamW (learning rate 10^{-3} , weight decay 10^{-4}), Fusion (F), and Unknown (Q) [9, 30]. The class distribution and the mapping are presented in Table 1. In accordance with recent literature [3], batch size 64, gradient-norm clipping at 1.0 and a cosine learning-rate schedule over a nominal 100-epoch budget. Early stopping monitors DS1-validation Macro-F1 with patience 15, and the checkpoint of the Q class is retained for reporting completeness but not used for weighting losses because of its small sample size ($< 0.02\%$). best validation epoch is retained. In the 60 class-balanced runs of the three PC-WavKAN configurations whose DS2 results are reported, training stopped between epochs 17 and 40, and at the retained epoch the learning rate had fallen by at most 15%. The baseline scripts did not log their stopping epoch.

Class Distribution and AAMI Standard Mapping. The imbalance ratio between Normal (N) and Fusion (F) beats is approximately 112:1, motivating the curriculum learning strategy described in Section 3.

The AdamW optimizer was used with an initial learning rate of 1×10^{-3} and weight decay of 1×10^{-4} to train the models. Early stopping was monitored on the validation Macro-F1 score with patience being 10 epochs. Training follows the two-phase curriculum schedule described in Section 3, i.e., Phase 1 (epochs 1-15) using only minority class samples along with standard

Class-balanced sampling. For the first 25 epochs, mini-batches are drawn by a sampler that selects each class with equal probability, and the loss is unweighted cross-entropy^{loss} and Phase 2 (epochs 16-50) using the full dataset along with . Each class therefore supplies about one fifth of every mini-batch, so the 6 Q and 28 F training beats are each redrawn many times per epoch. The training script then enters a second phase in which the sampler is annealed towards the natural class distribution and the loss becomes a class-weighted focal loss. In all 60 runs of the three configurations whose DS2 results are reported (20 seeds each), the retained checkpoint came from the first phase (best epochs 2–25), and 20 of these runs stopped before the switch. Of all 160 runs that used this schedule, including the five class-balanced ablation arms of Table 4,

Table 2: Training and evaluation settings. Every setting above the rule is shared by all five models; the settings below it differ by design (§3.5).

Setting	Value
Inputs	beat window (360 samples) and five RR intervals
Optimiser	AdamW; learning rate 10^{-3} ; weight decay 10^{-4}
Learning-rate schedule	cosine annealing over a nominal 100 epochs
Batch size; gradient clipping	64; gradient norm 1.0
Early stopping	patience of 15 epochs on DS1-validation Macro-F1
Retained checkpoint	epoch with the best DS1-validation Macro-F1
Seeds	20, identical for every model (§3.7)
Imbalance handling, PC-WavKAN	class-balanced sampling with unweighted cross-entropy
Imbalance handling, baselines	natural sampling; inverse-frequency weighted cross-entropy (S weight $\times 8$); focal loss without weights for CNN+Focal
Augmentation	PC-WavKAN only: non-N beats (noise, baseline wander, scaling)

158 retained a first-phase checkpoint; the other two are ablation runs that retained epoch 26 or 27 (Morlet basis, seed 333; no side branch, seed 13). The second phase therefore did not contribute to any model whose DS2 result is reported. We describe and evaluate the procedure by what determined those models: class-balanced sampling with unweighted cross-entropy and validation-based checkpoint selection, abbreviated CBS. The comparator, used by three of the four baselines and by the no-CBS arms, is natural-distribution sampling with inverse-frequency class-weighted cross-entropy loss and $10\times$, with the S-class weight emphasis [17]. weight multiplied by 8. CNN+Focal uses natural-distribution sampling with focal loss and no class weights.

To test the contribution of the WavKAN-CL architecture and training strategy, we compare WavKAN-CL to three internal baselines (Table 3). These comparisons separate the effect of morphological feature extraction (CNN vs.

Augmentation. During training only, every non-N beat in a mini-batch is perturbed with probability one. The perturbation is additive Gaussian noise at a 25 dB signal-to-noise ratio, a sinusoidal baseline wander of 0.5–2.5 Hz with amplitude 0.01–0.05 (in z -units), and amplitude scaling by $\mathcal{U}(0.9, 1.1)$, applied after z -scoring. The baselines are trained without augmentation (§3.5).

3.5. Baselines and What Is Matched

Four architectures spanning 30K–118K parameters are compared (Table 3). *Held identical:* data, preprocessing, split, input representation (every model receives both the beat window and the interval vector), optimiser, learning rate, weight decay, schedule, epoch budget, early-stopping rule and patience, batch size, gradient clipping, checkpoint-selection rule, the 20 seeds, and, wherever class weights are used, the class-weight formula. *Not held identical:* the imbalance mechanism and augmentation. The baselines use natural-distribution sampling with class-weighted cross-entropy, except CNN+Focal, which uses focal loss without static class weights. We also tested, for the baselines, a class-balanced sampler combined with an inverse-frequency-weighted focal loss. At this imbalance ratio that combination drove the majority-class gradient to near zero, and three of the four baselines then never predicted class N, so it was not used; those runs were not retained. The comparison in §4.2 is therefore between complete methods, i.e. WavKAN), rhythm integration and curriculum learning. architecture together with its imbalance strategy, under a common protocol and budget. §4.4 and §4.5 quantify the two mechanisms on PC-WavKAN itself.

Baseline Model Definitions. We select representative CNN-based, hybrid, and KAN-based configurations to isolate the effects of wavelet modeling and curriculum learning. The baseline labelled B-Spline KAN is the same

wavelet-KAN layer with a single cubic B-spline in place of the Mexican Hat on every edge, isotropic initialisation, no side branch, no dropout after the layer normalisation, and the same GRU, rhythm MLP and head. It is not the grid-based spline KAN of Liu *et al.* [12].

Table 3: Baselines. Protocol, split, inputs, optimiser, budget, checkpoint rule and seeds are identical to PC-WavKAN (§3.5).

Model	Params	Imbalance handling
ResNet1D (4-block residual CNN, 32 channels)	62,869	Weighted cross-entropy
Transformer ($d_{\text{model}}=32$, 2 layers, 4 heads)	29,653	Weighted cross-entropy
CNN+Focal (4-layer 1D CNN)	71,061	Focal loss ($\gamma=2$)
B-Spline KAN (single B-spline per edge; no PCWI or side branch)	118,229	Weighted cross-entropy
PC-WavKAN (§3.3)	153,045	CBS + augmentation

Our ablation experiments reveal three critical findings regarding the WavKAN-CL framework:

Curriculum Impact: The proposed curriculum learning strategy prevents the majority N-class from dominating gradient updates during early training

3.6. Parameter-Space Prior Retention

To test RQ3 we measure how close the trained wavelet parameters remain to the PCWI prior values. For block c ,

$$\text{PPR}_c = \max\left(0, 1 - \frac{1}{2} \left[\frac{|\overline{\mu_{j,k}} - \bar{\mu}_c|}{\Delta_\mu} + \frac{|\overline{|\gamma'_{j,k}|} - \bar{\gamma}_c|}{\Delta_\gamma} \right] \right), \quad (4)$$

where the overlines denote means over all edges of block c , and $\Delta_\mu = 0.40$, $\Delta_\gamma = 0.20$ are fixed normalising spans. The reported PPR is the unweighted mean over the three blocks. The measure is computed from parameters alone and never observes a model output. The analysis is therefore at the level of basis-function parameters only. It makes no claim about features, attributions or individual predictions, which also pass through the GRU, the side branch, the rhythm branch and the classifier head. Because PCWI initialises at the prior values, an untrained PCWI model scores near-maximally by construction. PPR is therefore interpretable only against reference points, and we report three: untrained with PCWI, untrained with isotropic initialisation, and *trained* with isotropic initialisation. The last one tests whether training moves parameters towards the prior values. Per-edge absolute deviations are averaged rather than block means, because per-edge drift is close to symmetric and largely cancels in the mean (in the trained model, mean per-edge $|\Delta_\mu| = 0.064$ against 0.004 for the deviation of the block means). As Table ?? shows, this reduces the validation-test divergence ($0.06 \rightarrow 0.00$)

3.7. Statistical Methodology

Every arm is trained with the same 20 seeds: 42, 101, suggesting improved alignment between validation and test performance under strict inter-patient evaluation. **Statistical Rigor (p-values):** In order to determine if the performance increases were due to random initialization, we conducted a 777, 2026, 9999, 1234, 2024, 31415, 27182, 7, 11, 13, 888, 555, 333, 99, 1001, 5050, 8080 and 1998. Comparisons are paired by seed label. Within one architecture, the same seed gives the same initialisation. Across architectures the pairing is by label only and implies no shared initialisation or data order. The signed-rank test remains valid in that case under the null hypothesis of identical distributions. Significance is assessed by the two-sided Wilcoxon signed-rank test with 20 random seeds (Table ??). The difference in the Macro-F1 scores between the baseline and WavKAN-CL was found to be [31], with Holm–Bonferroni correction [32] within declared families. The metrics of one arm-versus-arm

comparison form one family, as do the comparisons against the four baselines on one metric, the seven variant-versus-base comparisons of the component ablation, and the five masked positions of the interval ablation. All p -values are Holm-adjusted unless labelled raw. Effect sizes are the paired standardised mean difference $d_z = \bar{\delta}/s_{\delta}$, reported against Cohen’s conventional thresholds [33]: $|d_z| < 0.2$ negligible (negl.), from 0.2 small, from 0.5 medium and from 0.8 large. **Positive d_z favours PC-WavKAN, the adopted configuration, or the treatment arm.** We also report 95% confidence intervals on the paired mean difference. The unit of replication is the training seed and the DS2 records are fixed, so the tests and intervals do not include the sampling variability of the 22 test patients. At $n = 20$ and $\alpha = 0.05$ two-sided, a paired t -test has 80% power only for $d_z \gtrsim 0.66$ (slightly more is needed for the signed-rank test), and non-significant at the 0.05 level ($p = 0.17$). This is an expected trade-off, since curriculum learning intentionally sacrifices some majority-class (N) precision in order to prioritize minority classes. Crucially, however, the curriculum strategy had a significant effect on mean V-recall ($0.871 \rightarrow 0.898$, $p < 0.05$) and reduced its variance ($\pm 0.021 \rightarrow \pm 0.011$), yielding more reliable ventricular detection rather than a single best-case performance. **RR-Interval Contribution:** A leave-one-out ablation study (detailed in Section 5) reveals that the $t - 3$ interval contributes most significantly to the detection of S-class ($\Delta = -0.062$, $p < 0.05$), whereas proximal intervals ($t - 1$, $t - 2$) make little contribution. These findings validate the architectural choice of the 5-beat contextual window. This stability advantage is visualized in Figure ???. While both training strategies yield similar mean results are reported with their intervals rather than as evidence of absence. No equivalence margin was specified, so parity results are not equivalence claims.

Outcome hierarchy. Results are reported in §4. The *primary* outcome is DS2 five-class Macro-F1 scores (≈ 0.36), of the adopted configuration against the four baselines. The *pre-specified secondary* outcome is the effect of the imbalance mechanism on per-class recall, tested on the base configuration (§4.4). Everything else is *exploratory*. That includes the same comparison on the adopted configuration, which uses arms trained for the augmentation analysis, and the interval ablation, the cross-database evaluation, the prior-retention analysis and the artefact sensitivity analysis. The last four were added or revised, and the four-class Macro-F1, a per-record S-recall breakdown and the curriculum-trained model exhibits a narrower IQR and eliminates the low-performance outliers seen in the baseline. This is clinically advantageous because a model with lower variance is more predictable in real-world deployment, reducing the risk of unexpected poor performance on unseen patient populations. record-202 sensitivity analysis were added, after the primary results were known. Exploratory analyses are labelled as such where reported.

In order to strictly justify the architectural decisions of WavKAN-CL, we performed a thorough ablation study over three main dimensions: choice of wavelet basis function, temporal integration module (GRU) and RR-interval fusion strategy (Table??). The learnable Mexican Hat basis was consistently found to outperform Morlet and DOG (Derivative of Gaussian) wavelets in achieving a high Ventricular recall with lower variance. Furthermore, removal of the bidirectional GRU (pure spatial processing) or the 5-beat rhythm branch resulted in substantial performance drops across all classes, validating the necessity of our dual-branch spatiotemporal architecture for accurate inter-patient generalization.

Software. Models are implemented in PyTorch. Every quantitative claim in this paper is re-derived from the per-seed result files by a verification script released with the code (Code availability).

Notably, the B-spline KAN baseline achieves higher point-estimate metrics (Macro-F1: 0.396, V-Recall: 0.920) than the proposed Mexican Hat configuration. However, this comparison must be interpreted within the broader objectives of WavKAN-CL. First, Bozorgasl & Chen [13] demonstrated that B-spline KANs suffer from performance degradation under grid refinement due to noise capture, whereas wavelet-based approximations inherently resist overfitting—a property reflected in our model’s near-zero validation–test gap (Table ??). Second, the Mexican Hat wavelet offers a physiologically grounded inductive bias: its second-derivative-of-Gaussian form naturally aligns with QRS complex morphology [15], enabling clinically interpretable learned representations (Fig. 4) that B-splines, as generic local polynomials, cannot provide. Third, within the XAI taxonomy of Taleban et al. [7], only wavelet-parameterized edges qualify as *intrinsically interpretable*, whereas B-spline edge functions remain opaque polynomial fits without domain-specific semantics. Thus,

4. Results

4.1. Architecture Selection on Validation Data

The configuration reported in the rest of this paper is the one that a rule using DS1 validation only selects; when that rule was applied relative to the DS2 results is disclosed under Test-set exposure below. Table 4 gives the single-variable ablation. The base configuration has the wavelet-KAN layer with PCWI, the selection of gated side branch, the Mexican Hat basis reflects a principled trade-off: accepting a modest reduction in aggregate metrics in exchange for structural interpretability, implicit regularization, and physiological grounding—properties that are essential for clinical trust and regulatory compliance in safety-critical biomedical applications. self-attention rhythm encoder and CBS, and each row changes exactly one component. The validation score of each seed is taken at the epoch its checkpoint was saved.

Comprehensive Architectural Ablation Study (DS2 Test Set). Models were evaluated across multiple random seeds. The proposed configuration (Mexican Hat, BiGRU, and RR-fusion) yields the highest and most stable V-class safety.

Table 4: Single-variable component ablation on **DS1 validation**, 20 seeds per row. Base configuration: PCWI, gated side branch, self-attention rhythm encoder and CBS; validation Macro-F1 0.4145 ± 0.0201 . Δ and d_z are variant minus base, so positive values favour the variant. p is Holm-adjusted across the seven comparisons. Removing the side branch also removes 34,816 parameters (22.6% of the base model). DOG: first derivative of Gaussian.

Variant	Val. Macro-F1	Δ vs base	d_z	Holm p	Reading
No PCWI (isotropic init.)	0.3932 ± 0.0129	-0.0213	-1.05 (large)	1×10^{-4}	PCWI helps in this configuration
No gated side branch	0.3976 ± 0.0184	-0.0169	-0.54 (medium)	0.164	not resolved
No CBS (reweighting)	0.4053 ± 0.0173	-0.0092	-0.37 (small)	0.624	not resolved
Basis: Morlet	0.4219 ± 0.0187	$+0.0074$	$+0.31$ (small)	0.568	not resolved
Basis: DOG	0.4041 ± 0.0123	-0.0104	-0.45 (small)	0.164	not resolved
Basis: B-spline	0.4116 ± 0.0151	-0.0029	-0.14 (negl.)	0.624	not resolved
MLP rhythm encoder	0.4397 ± 0.0140	$+0.0252$	$+1.15$ (large)	8×10^{-4}	adopted

Three findings follow. *Initialisation*: replacing PCWI with the isotropic initialisation costs 0.021 validation Macro-F1 ($d_z = -1.05$, Holm $p = 1 \times 10^{-4}$). It is the only component whose removal significantly reduced validation Macro-F1. The comparison contrasts one structured initialisation with one isotropic distribution, and §3.3 shows that what PCWI fixes is a set of amplitude offsets and widths, not a physiological assignment. The finding is therefore that this initialisation helps, not that physiological content does. It was measured in the base configuration and was not re-measured on the adopted configuration. *Mother wavelet*: substituting a Morlet wavelet, the first derivative of Gaussian (DOG) or a B-spline for the Mexican Hat changes validation Macro-F1 by at most 0.010, and no difference approaches significance. *Rhythm encoder*: replacing self-attention with an MLP over the same five intervals improves validation Macro-F1 by 0.025 ($d_z = +1.15$, Holm $p = 8 \times 10^{-4}$). This configuration is the best of all eight on validation. It outperforms each of the seven alternatives (Holm-adjusted $p \leq 0.0032$ over that family), and we adopt it.

Test-set exposure. The adopted configuration was first identified during development, after DS2 metrics of all eight configurations had been computed; at that point the base configuration was known to be significantly below each of the four baselines on DS2 (Holm-adjusted $p \leq 0.005$). The validation analysis of Table 4, carried out afterwards, selects the same configuration, and it is the only selection rule we report. The configuration ranked first on validation is also first of the eight on DS2 (0.357 ± 0.019 ; base configuration 0.323 ± 0.018). DS2 was therefore not untouched, and §4.2 should be read as a held-out estimate under a validation-based rule applied after DS2 had been inspected, not as a single-shot test. Any bias from this exposure would favour PC-WavKAN. On arrays regenerated from PhysioNet with the released

code, all 20 adopted-configuration checkpoints reproduce their DS2 predictions beat for beat, and 19 reproduce exactly the validation Macro-F1 logged at their saved epoch. Seed 1001's logged validation score does not reproduce. Excluding it changes no corrected conclusion, though the unadjusted interval against B-Spline KAN (Table 5) then narrowly excludes zero ($[-0.031, -0.0002]$).

4.2. Primary Comparison

Table 5: **Analysis of Generalization Gap** **Primary outcome.** DS2 five-class Macro-F1, PC-WavKAN versus four baselines, 20 paired seeds. The overfitting gap often seen in the inter-patient paradigms Δ , CI and d_z are PC-WavKAN minus baseline; p is substantially reduced by Curriculum Learning Holm-adjusted across the four comparisons.

Model	Params	Macro-F1	Δ	95% CI of Δ	d_z / Holm p
ResNet1D	62,869	0.351 ± 0.019	+0.006	$[-0.008, +0.020]$	+0.19 / 1.00
Transformer	29,653	0.353 ± 0.040	+0.004	$[-0.013, +0.022]$	+0.12 / 1.00
CNN+Focal	71,061	0.362 ± 0.015	-0.005	$[-0.017, +0.006]$	-0.22 / 1.00
B-Spline KAN	118,229	0.371 ± 0.024	-0.014	$[-0.029, +0.001]$	-0.43 / 0.36
PC-WavKAN	153,045	0.357 ± 0.019	—	—	—

Statistical Rigor (20-Seed Comparison). The curriculum strategy significantly stabilizes V-Recall ($p < 0.05$); the variance of safety-critical predictions is reduced. PC-WavKAN's DS2 Macro-F1 is not statistically distinguishable from any baseline (Table 5). All Holm-adjusted $p \geq 0.36$, all effects are negligible to small, and every 95% interval on the paired difference contains zero, the widest bound being 0.029. The intervals exclude Macro-F1 differences larger than 0.03 in either direction. They do not establish equivalence, for which no margin was set. PC-WavKAN is also the largest of the five models. Because the Q class contributes a near-constant zero (§4.3), we also report the four-class N/S/V/F Macro-F1 (exploratory): PC-WavKAN 0.446 ± 0.023 , ResNet1D 0.439 ± 0.023 , Transformer 0.441 ± 0.050 , CNN+Focal 0.453 ± 0.019 and B-Spline KAN 0.463 ± 0.030 , with all Holm-adjusted $p \geq 0.36$. The conclusion is unchanged. Removing record 202, the DS2 record from the same subject as training record 201 (§3.2), lowers every model's Macro-F1 by 0.002 to 0.006 (PC-WavKAN: $0.357 \rightarrow 0.354$), and the comparison is again unchanged: all Holm-adjusted $p \geq 0.25$, and every paired interval lies within ± 0.03 (exploratory).

Fig. 3 shows the structure behind these values, averaged over the 20 seeds. N and V separate well, each leaking under 6% into the other, with V-recall 0.898. S beats are split almost equally between V (0.40) and N (0.39), and most F beats are predicted as N (0.78).

Training Convergence: Baseline vs. Curriculum Learning. (a) Training loss, (b) validation Macro-F1, (c) Ventricular (V) recall, and (d) Supraventricular (S) recall across 50 epochs. The dashed vertical line indicates the phase transition for the curriculum (epoch 15), where training shifts from minority-only samples to the full dataset. The curriculum strategy has higher early V-recall and S-recall during the discovery phase, demonstrating the benefit of early minority exposure.

Table?? presents the best-seed performance on the strictly held-out DS2 test set. The model prioritizes clinical safety and achieves a V-class Recall of 0.858 (mean of

4.3. Per-Class Operating Points

Table 6 gives precision, recall and F1 per class over 20 seeds: 0.88, as reported in Table ??). Consistent with literature on non-augmented inter-patient schemes [8], the detection of Supraventricular (S)beats remains challenging (Recall 0.290) due to their morphological similarity to Normal beats and severe class scarcity. Fusion (F)beats show near-chance performance (Recall 0.028), which is attributable to the extreme lack of samples (only 388 test samples, 15 validation samples) and their inherently ambiguous morphology overlapping with both N and V classes (Figseeds, recomputed from stored per-seed predictions. The per-class F1 values average to the Macro-F1 of Table 5 for all five models. 3). Notably, Zhou et al. [18] also excluded the F class from their performance

Analysis of Seed Stability (n=20 seeds). Boxplot comparison of the Macro-F1 scores for 20 random seeds. Curriculum learning demonstrates greater stability (tighter IQR) over the baseline, which has a greater volatility. Dashed lines represent mean, solid orange lines represent median.

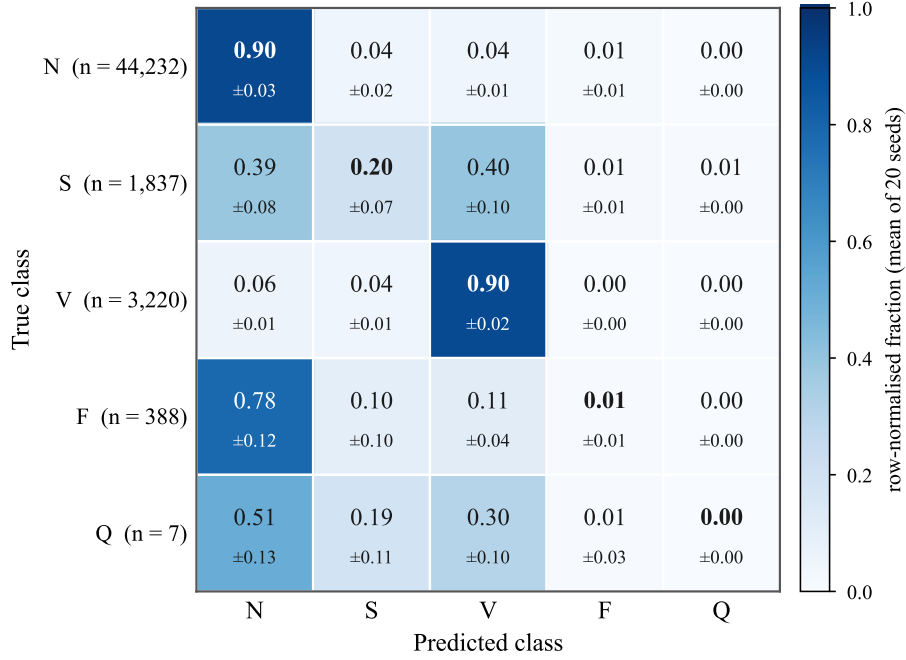


Figure 3: Row-normalised DS2 confusion matrix of PC-WavKAN, averaged over the 20 seeds. Each seed's matrix is normalised by row and then averaged, and the across-seed standard deviation is shown under each value. Rows are labelled with their support; the Q row rests on 7 beats. The diagonal equals the per-class recall of Table 6.

Table 6: Complete per Per-class precision, recall and F1, 20 seeds, DS2. (a) PC-WavKAN, all classes. (b) S class metrics across models, with the Holm-adjusted paired comparison of S-F1 against PC-WavKAN (Test Set DS2exploratory; positive d_z favours PC-WavKAN). Note the good Recall for the life-threatening Ventricular S-class differences between models combine architecture and imbalance mechanism (V§3.5)class. Q scores exactly zero for every model.

(a) PC-WavKAN, per class				
Class	Support	Precision	Recall	F1
N	44,232	0.971 ± 0.004	0.905 ± 0.025	0.936 ± 0.013
S	1,837	0.175 ± 0.079	0.198 ± 0.067	0.180 ± 0.064
V	3,220	0.527 ± 0.051	0.898 ± 0.020	0.663 ± 0.039
F	388	0.005 ± 0.006	0.006 ± 0.006	0.005 ± 0.006
Q	7	0.000	0.000	0.000
(b) S class across models				
Model	S-Prec.	S-Rec.	S-F1	d_z / Holm p
PC-WavKAN	0.175	0.198	0.180	—
ResNet1D	0.173	0.100	0.124	+0.77 / 0.004
Transformer	0.219	0.229	0.207	−0.35 / 0.312
CNN+Focal	0.306	0.075	0.115	+0.85 / 0.002
B-Spline KAN	0.237	0.280	0.244	−0.66 / 0.027

analysis because it represented less than 1% of training data, confirming that this extreme scarcity is a recognized open challenge rather than a model-specific failure.

Normalized Confusion Matrix (DS2 Test Set). The model demonstrates high recall for the life-threatening Ventricular (V) class (0.86) with stable Normal (N) classification (0.82). Confusion between S and N classes is an expression of morphological similarity of the supraventricular beats to normal sinus rhythm under the strict inter-patient evaluation. To evaluate generalization to unseen recording environments, the model trained on MIT-BIH was evaluated zero-shot on the PTB-XL test set (Table ??). While overall macro-averaged performance declined under this severe domain shift, the network was able to maintain a measurable sensitivity to Ventricular (V) The S-class operating point is poor in both directions. Recall is 0.198 at precision 0.175, so roughly four in five predicted S beats are wrong and four in five true S beats are missed. On S-F1, PC-WavKAN is better than ResNet1D and CNN+Focal, indistinguishable from the Transformer, and worse than B-Spline KAN ($d_z = -0.66$, Holm $p = 0.027$). Its S-recall advantage over ResNet1D and CNN+Focal comes largely from record 232, where those two models detect almost no S beats (Limitation 2). The parity of §4.2 is therefore not uniform across classes. Q contributes an exact zero for every model, consistent with its 6 training beats. With Q at zero, five-class Macro-F1 cannot exceed 0.8, which should be recalled whenever five-class inter-patient Macro-F1 values are compared.

Comparison with the protocol’s source. de Chazal *et al.* [11], whose partition we adopt, report a sensitivity of 0.759 at positive predictivity 0.385 for supraventricular ectopic beats (0.108 Recall, 5,889 support) and Normal beats (0.729 Recall, 53,497 support). SVEB, our S class), and 0.777 at 0.819 for ventricular ectopic beats (VEB, our V class). On the S class their linear-discriminant system is substantially better than every model trained here on both axes. On the V class our recall of 0.898 exceeds theirs, while our precision of 0.527 is lower. Their system uses two leads, hand-engineered morphology at fiducial points and RR-interval features, where ours uses one lead and a fixed window. We have not rerun their method. We report the comparison because the S-class gap is large and is against the protocol’s own source.

Zero-Shot Cross-Dataset Performance on PTB-XL Test Set. Results were obtained using the model only trained on MIT-BIH (DS1) without any target domain fine-tuning.

4.4. Class-Balanced Sampling Versus Reweighting

The pre-specified test compares CBS with reweighting on the base configuration, with augmentation in both arms. S-recall rises from 0.079 ± 0.016 to 0.122 ± 0.052 (Holm $p = 0.0016$ across five metrics, $d_z = +0.77$). On DS1 validation it rises from 0.218 ± 0.048 to 0.305 ± 0.047 ($d_z = +1.36$, raw $p = 2.4 \times 10^{-4}$). Table 7 repeats the comparison on the adopted configuration. It uses the two augmentation-free arms trained for §4.5, so the analysis is exploratory.

Table 7: CBS versus reweighting on the adopted configuration, both arms without augmentation, 20 paired seeds (exploratory). DS2 rows: d_z and p are CBS minus reweighting, Holm-adjusted across the five metrics. Validation rows: values at each run’s retained epoch, raw p .

Metric	Reweighting	CBS	d_z	p
<i>DS2 (held out)</i>				
Macro-F1	0.351 ± 0.014	0.364 ± 0.021	+0.52 (medium)	0.098
S-Recall	0.137 ± 0.037	0.222 ± 0.057	+1.47 (large)	2.9×10^{-5}
F-Recall	0.0003 ± 0.0008	0.0022 ± 0.0019	+0.88 (large)	0.013
V-Recall	0.880 ± 0.024	0.880 ± 0.017	0.00 (negl.)	0.756
N-Recall	0.927 ± 0.021	0.917 ± 0.024	−0.32 (small)	0.228
<i>DS1 validation</i>				
Macro-F1	0.4285 ± 0.0125	0.4195 ± 0.0098	−0.57 (medium)	0.027 (raw)
S-Recall	0.2639 ± 0.0510	0.2391 ± 0.0425	−0.53 (medium)	0.025 (raw)

A comparison between WavKAN-CL and recent studies on the MIT-BIH is found in Table ?? . A key methodological distinction must be noted: MAK-Net [1] and several prior methods employ an intra-patient evaluation protocol (random beat-level train/test splits), which inflates reported metrics through patient-specific memorization [8]. In fact, a recent systematic review of 122 studies by Silva et al. [4] found that intra-patient models continue to be highly over-represented in the citation metrics of the literature, potentially creating a misleading perception of the state-of-the-art. Even recent lightweight architectures designed for edge deployment, such as Elsheimy et al. [10] (99.18% accuracy, CNN+Attention), would still rely upon intra-patient random splits (60/20/20) to obtain their results, which are therefore not directly comparable to inter-patient results. WavKAN-CL, by contrast, is evaluated under the strict inter-patient DS1/On DS2 protocol where all test patients are totally unseen during training. Critically, although Zhao et al. [1] showed that SMOTE can significantly improve performance when intra-patient evaluation was performed (accuracy 0.998), our experiments in strict inter-patient evaluations show the opposite: SMOTE-based temporal interpolation generates physiologically unrealistic representations, reducing S-Recall , CBS raises S-recall from 0.290 to 0.245—a finding that underscores the importance of protocol choice for evaluating augmentation strategies. Despite using this more stringent protocol and operating with only 95, 189 parameters (0.38 MB), WavKAN-CL has a V-Recall of 0.86 with a median inference latency of 0.78 ms per beat—over $30\times$ faster than MAK-Net (24 ms per beat, 6.1 M parameters)[1], and well within the requirements of edge-deployable Green AI [10].

0.137 to 0.222, with a 95% confidence interval of $[+0.058, +0.112]$ on the paired difference. The absolute F-recall gain is negligible ($0.0003 \rightarrow 0.0022$). V- and N-recall show no corrected change. On the validation partition the direction reverses. CBS is nominally worse on both validation Macro-F1 and S-recall, although after Holm correction across the two validation metrics neither difference is significant ($p \approx 0.05$). The DS2 gain in S-recall therefore replicates in two configurations but is not reproduced on the four-record validation partition for the adopted configuration. We read this as indicating that the benefit of class-balanced sampling for supraventricular beats may depend on the patients evaluated, and we do not claim it as a general property. The pre-specified test on the base configuration is the confirmatory evidence, and it is modest in absolute terms: an S-recall of 0.122.

Comparison with Recent Studies on MIT-BIH Arrhythmia Database. [†]Evaluated under intra-patient protocol (random beat-level split); not directly comparable to inter-patient DS1/DS2 results. WavKAN-CL is the only structurally interpretable model evaluated under the strict inter-patient protocol.

The model achieves good Ventricular (V) class recall (0.88 ± 0.03 across 20 seeds), which is the primary safety metric in clinical arrhythmia monitoring. However, the global

4.5. Augmentation

PC-WavKAN is trained with minority-class augmentation and the baselines are not; this analysis is exploratory. On validation, augmentation is supported: with CBS in both arms, it raises validation Macro-F1 score of 0.362 shows that the model does not yet perform equally well across all AAMI superclasses, highlighting a key limitation. As systematically reviewed by Xiao et al. [9], the transition from intra-patient to strict inter-patient evaluation usually results in a drop of F1 scores of over 15% due to inter-subject variability. Our Macro-F1 is consistent with by 0.020 ($d_z = +1.16$, raw $p = 1.7 \times 10^{-4}$), consistent with its place in the benchmarks reported by Bahrami & Fotouhi [8], who showed that unaugmented inter-patient models achieve adopted recipe. On DS2, removing it leaves Macro-F1 scores within the 0.35 – 0.40 range. Importantly, when evaluated against the E3C evaluation criteria proposed by Silva et al. [4]—which require (i) strict inter-patient evaluation, statistically unchanged (0.357 with augmentation and 0.364 without; Holm $p = 0.46$ across four metrics) and reduces F-recall (ii) class-specific metrics for imbalanced data, 0.0061 to 0.0022, Holm $p = 0.043$). The augmentation asymmetry therefore does not explain the parity in Table 5. A separate, shorter study used a different recipe: 60 epochs with patience 12, class-balanced sampling combined with class-weighted cross-entropy throughout, and noise settings that differ from the adopted augmentation. In it, the three noise-based strategies raised V-recall and lowered Macro-F1, and a SMOTE-style interpolation [34], which replaces each minority beat by a random convex combination of it and a random beat of the same class in the mini-batch (possibly itself), produced the largest S-recall gain ($0.316 \rightarrow 0.648$) at the largest Macro-F1 cost. The Macro-F1 penalty of the noise-based strategies did not carry over to the full recipe, which cautions against extrapolating short single-mechanism ablations.

4.6. Interval Features, Lookahead and an Annotation-Derived Artefact

An exploratory leave-one-out analysis set each interval position in turn to 0.8 s at inference. It used the base configuration, whose rhythm features are identical to the adopted one’s, with 20 seeds, two-sided tests and Holm correction across the five positions. Masking RR_0 caused by far the largest loss of S-recall (iii) adherence to AAMI clinical protocols, and (iv) consideration of computational complexity—WavKAN-CL satisfies all four criteria, placing it among the mere 4.1% (5 out of 122) of recent ECG classification studies that meet these rigorous standards. Furthermore, Silva et al. stated that 60% of these fully compliant models require the use of Spiking Neural Networks (SNNs) or Custom FPGA Accelerators to achieve embedded feasibility. By contrast, WavKAN-CL achieves a 95K-parameter footprint and sub-millisecond inference using standard hardware, offering an accessible alternative for Green AI without requiring specialized neuromorphic chips. -0.0323 ± 0.0492 , Holm $p = 6 \times 10^{-4}$). This is expected, since an ectopic beat that fires early shortens its preceding interval, the classical sign of prematurity. Masking RR_{+2} caused a small loss that survives correction (-0.0045 ± 0.0103 , Holm $p = 0.028$). The other three positions had no corrected effect. The model therefore draws signal from an interval that has not elapsed when beat i occurs, which is one reason we treat the task as offline (§3.1).

Nevertheless, we do not claim overall classification superiority on this metric; the primary contribution of WavKAN-CL lies in achieving competitive V-class safety with $> 95\%$ fewer parameters than existing baselines. A stabilizing effect of curriculum learning is demonstrated by the low validation-test divergence (Gap ≤ 0.01 across

Annotation-derived artefact. Because the intervals are computed between consecutive annotation-file entries, including non-beat markers, 4,578 of 49,684 DS2 beats (9.2%) have at least one interval that differs from the beat-to-beat value, and 1,433 (2.9%) have a different RR_0 . By class, RR_0 differs for 6.4% of V beats, against 2.7% of N beats and 1.3% of S beats, and the recorded value is almost always shorter than the true interval, which mimics prematurity. A rhythm-change marker is placed where an arrhythmic episode begins, so this artefact carries annotation-derived information that would not be available on unannotated recordings, and it correlates with the label. We quantified its effect without retraining, as an exploratory analysis. We recomputed every DS2 interval between beat annotations only and re-evaluated all 100 published checkpoints (five models, 20 seeds). To confirm generalization beyond the MIT-BIH dataset, we performed a strict zero-shot cross-dataset evaluation on the PTB-XL database (The recomputation leaves the beats and their order unchanged. As a check, the same re-evaluation with the published intervals reproduced every model’s reported Macro-F1. Of the 100 Hz, 21 checkpoints, 88 reproduced every DS2 prediction exactly. The remaining 12, all CNN+Focal, differed in 1 to 4 of 49,837 patients). As expected under severe domain shift with no fine-tuning, total precision declined when compared to intra-dataset benchmarks but the model maintained a Normal (N) recall of 0.729. This large drop in sensitivity to minority classes without target-domain adaptation highlights the large morphological variability between the two recording environments and strengthens the known difficulties for the classification of ECG arrhythmia with cross-dataset classification 684 beats, which we attribute to floating-point differences between the original GPU inference and the re-evaluation (TF32 disabled; deterministic algorithms not enforced).

Confusion persists regarding Normal (N) and Supraventricular (S) beats particularly during early training stages of the curriculum. This presumably occurs due to subtle morphological deviations in S-beats that are very similar to Normal beats, especially in the absence of distinctive P-wave abnormalities. As mentioned by Zhou et al. [18], learning a robust decision boundary for the minority S-class (less than 2% of data) remains a primary challenge of inter-patient schemes. To directly address this class scarcity, we conducted a secondary experiment by using the Synthetic Minority Oversampling Technique (SMOTE) in the raw temporal feature space to even the representation of the minority classes before the curriculum learning. Interestingly, empirical results showed that SMOTE did not improve generalizability on the strictly held out inter-patient test set (S-Recall). With beat-to-beat intervals, the Macro-F1 of every model rose slightly, in all 20 seeds. The mean increase was $+0.002$ to $+0.005$ across the five models (PC-WavKAN: 0.245 vs. baseline 0.290; V-Recall: 0.886 vs. baseline 0.858). This negative result is consistent with the findings of Bae & Shin [23],

who demonstrated that naive augmentation methods degrade performance on imbalanced medical datasets. Schmale et al. [17] reported an analogous phenomenon in 12-lead ECG classification, where simulated signals caused misclassification between morphologically similar pathologies (LBBB vs. RBBB), confirming that synthetic representations which fail to capture subtle inter-class distinctions can confound classifiers. In contrast, in Zhou et al. [18] the improved SVEB recognition was achieved by augmentation using additive random noise (low-frequency, high-frequency, and white noise) rather than interpolation, without losing the physiological structure of the underlying beats. Together, these results highlight a critical limitation of spatial-interpolation techniques in time-series ECG data: synthetic interpolation between morphologically diverse arrhythmic beats often generates physiologically unrealistic representations that confound the classifier. Thus, our unaugmented Curriculum Learning strategy is the strongest for actual inter-patient generalization. $0.357 \rightarrow 0.360$), and PC-WavKAN’s V-recall changed by -0.001 . The primary comparison was unchanged: all Holm-adjusted $p \geq 0.38$, and every paired interval still includes zero. The artefact therefore did not inflate the reported performance, and removing it at inference does not change the comparative conclusions. The models were trained with it, and whether training on beat-to-beat intervals would change them was not tested.

4.7. Retention of the Prior Values

Table 8 evaluates PPR against its three reference points (exploratory).

In order to quantify the contribution of each of the rhythm features, we performed a leave-one-out ablation study on the 5-element RR-interval history vector. For each position ($t - 5$

Table 8: Parameter-space Prior Retention (4), mean $\pm$ std over 20 models, with mean absolute per-edge deviations from the prior values. Untrained rows use 20 initialisation seeds. The trained isotropic row uses the 20 checkpoints of the no-PCWI ablation arm (base configuration).

Model state	PPR	$ \Delta\mu $	$ \Delta\gamma $
Untrained, PCWI	0.9729 ± 0.0001	0.013	0.004
Trained, PCWI (PC-WavKAN)	0.7826 ± 0.0351	0.064	0.055
Untrained, isotropic	0.6698 ± 0.0008	0.161	0.051
Trained, isotropic	0.5794 ± 0.0173	0.177	0.080

Training moves the PCWI parameters substantially: PPR falls from 0.973 to $t - 1$), the corresponding feature was masked (set to zero) and 0.783, and the change in S-class recall was measured. Results of mean per-edge translation deviation grows fivefold. The trained PCWI basis nonetheless remains closer to the prior values than an untrained isotropic basis (0.783 against 0.670). The trained isotropic row decides the interpretation. When training starts from the isotropic initialisation, it moves the parameters *further* from the prior values ($0.670 \rightarrow 0.579$). Both comparisons separate the 20 random seed averages are presented in Fig.?? and show a statistically significant and physiologically interpretable pattern. Removal of the $t - 3$ interval (three beats prior) led to a consistent and statistically significant decrease in models completely (two-sided Mann–Whitney test, raw $p = 7 \times 10^{-8}$ by the normal approximation). Training from the isotropic initialisation therefore does not lead towards the prior values. The trained PCWI model’s proximity to them reflects where it started, not a learned preference. Together with the exchangeability of the channel blocks (§3.3), this means the wavelet parameters of this model do not support a physiological interpretation. They support the modest statement that the initial structure is partly retained. Fig. 4 shows the per-edge parameters behind these values. The retention lies mainly in the translations, which stay centred on the prior values. The dilations of the P- and T-prior blocks spread to a distribution broadly similar to that of the isotropic arm. Automated interpretability metrics for sparse autoencoders can score randomly initialised transformers similarly to trained ones [35]. For B-spline KANs, an untrained random-initialisation control has shown one edge-function

metric to be imposed by the basis rather than learned [36], and a weight-randomisation check has been applied to KAN attributions [37]. Our results indicate that parameter-space scores for KANs need the same reference points.

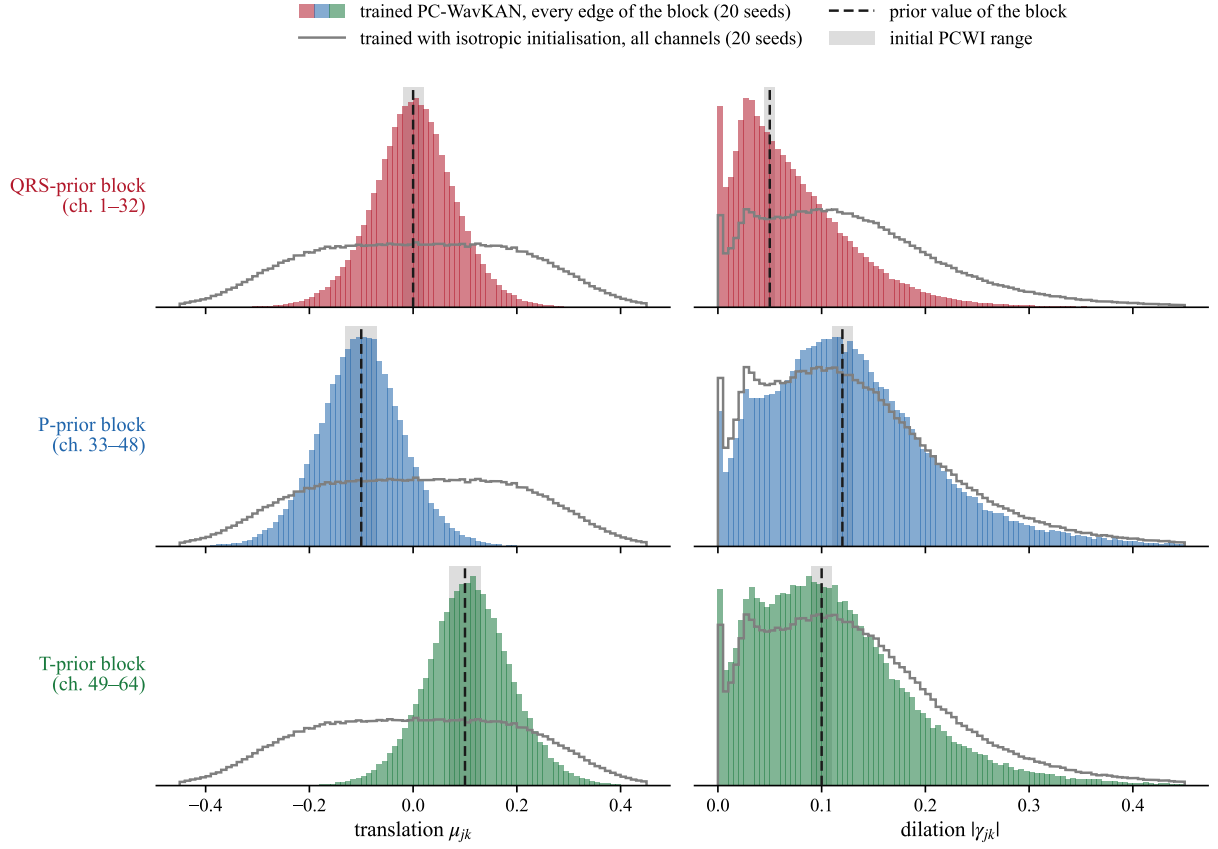


Figure 4: Per-edge wavelet parameters after training. Coloured histograms pool every edge of each channel block over all 20 seeds of PC-WavKAN. They are shown against the block’s prior value (dashed line) and initial PCWI range (grey band), and against the trained isotropic-initialisation arm of the base configuration, all channels (grey outline; the trained null of Table 8). Histograms are normalised to unit area. Translations stay centred on the prior values but spread well beyond their initial range. The dilations of the P- and T-prior blocks spread to a distribution broadly similar to the null’s. Block names refer to initial values, not to functions.

4.8. Cross-Database Evaluation

Table 9 evaluates the five models trained on MIT-BIH on two further databases, forward pass only, 20 seeds each. The first is the St Petersburg INCART 12-lead Arrhythmia Database [28, 29], all 75 records, lead II. The second is the MIT-BIH Supraventricular Arrhythmia Database (SVDB) [38], 78 records, channel 0, whose leads are not labelled. Both were resampled to 360 Hz and then passed through a re-implementation of the training extraction. A regression test confirms that it gives output byte-identical to the training extraction on MIT-BIH records, and the DS2 arrays it regenerates are identical to the training-pipeline arrays. Neither database was used for any training, tuning or selection decision. SVDB was assembled to enrich supraventricular ectopy and serves as an S-class recall ($\Delta = -0.062, p < 0.05$), whereas proximal intervals ($t - 1, t - 2$) made negligible contribution. This observation suggests that mid-range short-term rhythm context plays a significant role in supraventricular arrhythmia discrimination stress test. The analysis is exploratory.

PC-WavKAN shows no transfer advantage. On INCART its Macro-F1 of 0.390 is significantly below ResNet1D and CNN+Focal (both 0.431; $d_z \leq -1.53$, Holm $p < 0.001$).

Table 9: Zero-shot cross-database evaluation under the training preprocessing, 20 seeds per model. d_z and p compare PC-WavKAN with each baseline on Macro-F1 (positive favours PC-WavKAN), Holm-adjusted within each database; bold marks Holm $p < 0.05$.

Model	INCART (175,785 beats)				SVDB (184,486 beats)			
	Macro-F1	d_z / Holm p	V-Rec.	S-Rec.	Macro-F1	d_z / Holm p	V-Rec.	S-Rec.
PC-WavKAN	0.390 $\pm$ 0.017	—	0.832	0.671	0.275 $\pm$ 0.012	—	0.788	0.126
ResNet1D	0.431 $\pm$ 0.017	−1.80 / < 0.001	0.849	0.760	0.350 $\pm$ 0.026	−2.49 / < 0.001	0.832	0.256
Transformer	0.398 $\pm$ 0.032	−0.24 / 0.245	0.852	0.817	0.316 $\pm$ 0.031	−1.36 / < 0.001	0.820	0.305
CNN+Focal	0.431 $\pm$ 0.020	−1.53 / < 0.001	0.807	0.560	0.354 $\pm$ 0.013	−4.86 / < 0.001	0.769	0.179
B-Spline KAN	0.367 $\pm$ 0.017	+0.83 / 0.003	0.820	0.606	0.267 $\pm$ 0.016	+0.33 / 0.231	0.791	0.125

It is indistinguishable from the Transformer and above B-Spline KAN ($d_z = +0.83$, Holm $p = 0.003$). On SVDB it is significantly below all three non-KAN baselines (Holm $p < 0.001$) and indistinguishable from B-Spline KAN, and the two KAN-family models have the lowest Macro-F1 of the five. Fig. 5 places the in-distribution and cross-database comparisons on one scale.

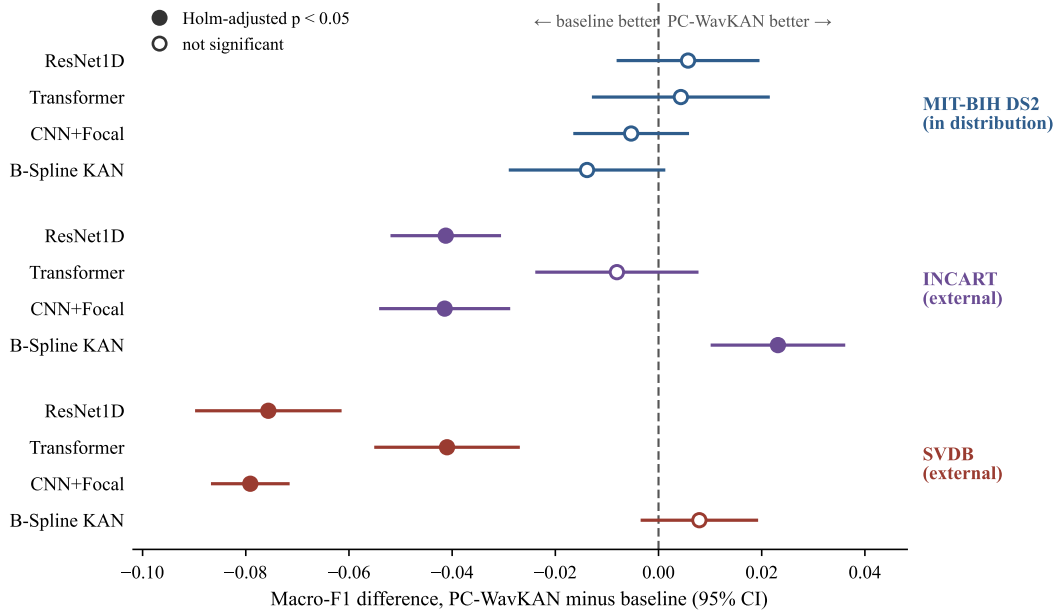


Figure 5: Paired Macro-F1 differences, PC-WavKAN minus each baseline, with 95% confidence intervals over 20 seeds. The three groups cover MIT-BIH DS2 (Table 5) and the two external databases (Table 9). Filled markers indicate Holm-adjusted $p < 0.05$ across the four baselines within a database; the dashed line marks zero. In distribution, every interval includes zero. Out of distribution, PC-WavKAN is below both convolutional baselines.

On SVDB, both KAN-family models also have the highest $S \rightarrow V$ confusion rates (0.431 and 0.451, mean over seeds, against 0.231–0.395 for the other models) and assign the largest share of beats to V (18.7% and 20.7%, against 8.4–17.6%). We report this as a descriptive pattern; the study does not identify its cause.

From a physiological perspective, this aligns with pre-ectopic compensatory patterns in case of PACs where rhythm irregularity is often reflected in earlier coupling intervals rather than the immediately preceding beats. The dominance of the $t - 3$ interval shows an empirical support for the 5-beat contextual design. It goes on to show that

Sensitivity to preprocessing. An earlier evaluation of the same checkpoints used windows from only 30 INCART records, on lead I, and applied no filter. It ranked PC-WavKAN first on INCART (Macro-F1 0.374 against 0.316–0.365 for the baselines). Matching the preprocessing

to the training data reversed that ranking. Cross-database comparisons of this kind are therefore sensitive to preprocessing details that are not always reported, and we report only the matched evaluation.

4.9. Efficiency

Table 10 reports inference on an x86 CPU for the seed-42 checkpoint, on random inputs and with PyTorch’s default thread count. Values are the mean of three repeated runs, each of 1000 timed trials at batch 1 (200 at batch 64) after a 20-iteration warm-up, and $\pm$ is the standard deviation across the three runs. The RR-branch is capturing clinically meaningful temporal dependencies and not a redundant auxiliary component. This is independently validated by Zhou et al. [18], who showed that adding RR interval features to their multiscale CNN notably improved SVEB classification from near-chance to competitive levels, confirming that rhythm context is essential for resolving the N-S ambiguity under inter-patient evaluation.

Table 10: Leave-One-Out Ablation of RR-Interval Features. Measured inference, PC-WavKAN, x86 CPU, PyTorch 2.6. Each bar represents the change in S-class recall when the corresponding RR interval is masked. The largest decrease in S-class recall occurs when the $t - 3$ interval is masked ($\Delta = -0.062, p < 0.05$). Error bars are ± 1 standard deviation.

Metric	Value
Model size, FP32 / INT8 dynamic	0.596 / 0.551 MiB (1.08 $\times$)
Latency, batch 1, FP32	2.93 ± 0.38 ms/beat (median 2.78)
Latency, batch 1, INT8 dynamic	4.00 ± 0.35 ms/beat (0.73 $\times$, slower)
Latency, batch 64, FP32	0.429 ms/beat
Process resident memory after benchmarking	235 MiB

Dynamic INT8 quantisation brings no benefit and slows inference. It was applied to the nn.Linear modules, which hold 11.4% of the parameters; the custom wavelet-KAN layers and the GRU, which hold 77.5%, remain in FP32. The FP32 latency supports continuous offline processing on a general-purpose CPU. We made no ARM or on-device measurement and make no claim about wearable deployment.

Kolmogorov-Arnold Networks in Biomedical Signals. Recent work has explored KAN-based architectures for ECG classification [1]. However, most implementations rely on spline-based edge functions. Our work is differentiated by the use of the Learnable Mexican Hat Wavelets which are well suited to modeling transient QRS dynamics [15]. As stated by Bozorgasl & Chen [13], by replacing fixed activations with learnable wavelets the model provides intrinsic structural interpretability in contrast to the opaque representations of conventional CNNs and Transformers (Fig. 4). Crucially, Bozorgasl & Chen [13] also demonstrated that wavelet-based approximations maintain a balance between accurately representing data structure and avoiding overfitting to noise unlike B-splines, whose performance degrades with grid refinement due to noise capture. This property contributes to WavKAN-CL’s low validation–test gap (Table ??), suggesting that the wavelet basis acts as an implicit regularizer. Within the XAI taxonomy proposed by Taleban et al. [7], WavKAN-CL belongs to the *intrinsically interpretable* family—the only methodological category that embeds transparency directly within the model architecture rather than relying on post-hoc attribution methods such as LIME or Grad-CAM. As Table ?? shows, competing ECG classifiers evaluated under inter-patient constraints do not explicitly incorporate intrinsic structural interpretability.

5. Discussion

Interpretability: Learned Wavelet Bases. WavKAN learns adaptive Mexican Hat wavelets (blue curves) that align with specific morphological features.

What the study shows. Under a matched protocol with multiplicity control, the wavelet KAN is not statistically distinguishable from four standard architectures on Macro-F1, with every paired interval within ± 0.03 . It is not more accurate than a 30K-parameter Transformer, and we do not present the wavelet-KAN basis as an accuracy mechanism. Among its design choices, in the configuration in which components were ablated, initialising the wavelet parameters in a structured, prior-centred way helped relative to an isotropic initialisation, and the choice of

mother wavelet did not measurably matter. The inspectability argument for wavelet KANs does not transfer to a physiological reading of this model’s parameters. The channel blocks are exchangeable, and training moves unconstrained parameters away from the prior values, not towards them. An inspectable parameter is not thereby an interpretable one. A claim that such parameters encode physiology would need behavioural evidence, for example a selective loss of one class when one block’s functions are perturbed. The null models we report are the minimum needed to interpret a parameter-space score.

Lightweight and Edge Deployable ECG Models: The development of resource efficient classifiers for ECG is important for wearable health monitoring. Farag [3] has demonstrated that miniature matched-filter CNNs can be employed for competitive performance with substantially reduced computational cost (≈ 15 KB). Similarly, Elsheikhy et al. [10] demonstrated the need for lightweight architectures for edge deployment. Our work is in line with this Green AI paradigm. WavKAN-CL achieves a Ventricular (V) recall of 0.86—which is similar to deep baselines such as Guo et al. [2], but using only 95k parameters

Self-attention over interval features. Removing self-attention from the rhythm encoder improved validation Macro-F1 with a large effect, while §4.6 shows that the interval features carry real signal, concentrated in RR_0 . The deficit therefore appears to lie in the encoder rather than in the features. One structural property of the evaluated encoder is consistent with this. Every interval is embedded by the same affine map, and the attention block has no residual connection. Whatever the weights, each output token of the block is therefore an affine function of two attention-weighted averages of the five intervals, one per head. The value of RR_0 enters those averages only in proportion to the attention it receives, and otherwise acts only through the attention weights, which depend on interval values rather than positions because the encoder has no positional encoding. The MLP receives the five intervals directly. We did not test whether this property causes the loss, and we observed the loss in one architecture and one dataset.

This corresponds to a $> 95\%$ reduction in parameter count relative to conventional DenseNet or Transformer architectures, prioritizing ventricular event detection and deployment feasibility over marginal gains in minority-class sensitivity

The external evaluation qualifies the parity. Under preprocessing matched to training, the wavelet KAN transferred worse than both convolutional baselines on both external databases, and the two KAN-family models were the weakest on SVDB. In this study, therefore, the parameterisation that is inspectable in principle came with no accuracy advantage in distribution and a disadvantage out of distribution. Anyone choosing an architecture for data from other sources should weigh the second against the first. The earlier, unmatched evaluation that appeared to show the opposite on INCART is a concrete example of how a preprocessing mismatch can manufacture an apparent transfer advantage.

Implications. Four practices follow from these results. Wavelet KANs and other KAN variants should be compared with standard architectures under the inter-patient protocol, over repeated seeds, with paired and multiplicity-corrected tests; under those conditions this model showed no Macro-F1 advantage (§4.2). Parameter-space interpretability scores should be reported against untrained and trained null models, because here an untrained prior-centred model scored highest (Table 8). Preprocessing should be verified identical before databases are compared, since a mismatch reversed the INCART ranking (§4.8). Features derived from annotation files should be audited, because RR intervals computed over all annotation entries carried label-correlated information (§4.6).

Despite the contributions of WavKAN-CL, several limitations should be acknowledged. First, the overall

5.1. Limitations

(1) Parity, not advantage or equivalence. No equivalence margin was set. The 95% intervals are compatible only with Macro-F1 score (0.362) remains modest compared to intra-patient benchmarks, reflecting the well-documented difficulty of strict inter-patient generalization under severe class imbalance [8, 9]. Second, the detection of Supraventricular (S) beats (Recall: 0.290) and Fusion (F) beats (Recall: 0.028) remains a significant challenge. The morphological overlap between S-beats and Normal sinus rhythm, combined with the extreme scarcity of F-class samples (less than differences within roughly ± 0.03 , and they describe variation over training seeds on the fixed DS2 records, not over patients).

(2) Absolute minority-class performance. S-recall is 0.198 at precision 0.175. F-recall is below 0.01 for every model and configuration (PC-WavKAN: 0.006), reflecting 28 training beats. Q, with 6 training beats, supports no per-class conclusion and adds a near-constant zero to the five-class average. One DS2 record, 232, holds 1% of training data), limits the model’s discriminative power for these classes under non-augmented conditions. Third, the B-spline KAN baseline achieves higher aggregate metrics (Table ??), indicating that the interpretability gains of the Mexican Hat basis come at a measurable cost in raw discriminative performance. Fourth, the model was trained exclusively on the ,382 of the 1,837 S beats (75%), so the DS2 S-class results largely reflect a single patient. In an exploratory breakdown over 20 seeds, PC-WavKAN’s S-recall is 0.138 on record 232 and 0.380 on the 455 S beats of the other records; every model is lower on record 232, where ResNet1D and CNN+Focal detect almost none of the S beats (0.004 and 0.013).

(3) Selection history of the test set. The adopted configuration is the one a validation-based rule selects, but it was first adopted after the DS2 metrics of all eight configurations had been seen, and the rule was formalised afterwards.

(4) Scope of the component evidence. Every component ablation, including PCWI, was run on the base configuration. The adopted configuration differs from it only in the rhythm encoder, and no other component was re-ablated on it. We do not claim that PCWI improves the adopted model.

(5) Validation partition. Validation comprises four records, and one record holds 90% of DS1’s fusion beats. Selection decisions and the validation reversal in Table 7 should be read with that in mind.

(6) Non-causal inputs and an annotation-derived artefact. The inputs include two following intervals, a window that often contains the next beat, and a zero-phase record-level filter. The interval features also carry an annotation-derived artefact in 9.2% of DS2 beats. Its effect is bounded in §4.6. It was not removed by retraining.

(7) External data. Cross-database evidence rests on two databases. INCART is evaluated on lead II rather than MIT-BIH Arrhythmia Database (360 Hz, two-lead); generalization to multi-lead recordings or different sampling rates (e.g., ’s modified lead II, and SVDB’s leads are unlabelled. PTB-XL at 100 Hz) was only evaluated in a zero-shot setting without domain adaptation, yielding limited cross-dataset transfer. Fifth, while the 95K-parameter footprint is edge-feasible, actual deployment on specific hardware platforms (e.g., ARM Cortex-M microcontrollers) has not yet been validated with post-training quantization and on-device latency benchmarking. [39] was processed but is excluded. Its labels are record-level, not beat-level. Our extraction used a symmetric window with the R-peak at sample 180 rather than 90, no filter, R-peaks from a peak detector rather than expert annotations, and interval features expressed as ratios to a local 10-beat mean rather than in seconds. A Macro-F1 of 0.200 ± 0.005 was obtained, and we attribute it to these mismatches rather than to transfer.

Future work will address these limitations through: (i) P-wave specific attention mechanisms for improved S-class discrimination, (ii) noise-based augmentation strategies that preserve physiological structure [18], (iii) multi-dataset training with domain adaptation, and (iv) INT8 quantization and deployment validation on low-power edge hardware **(8) Baselines and hardware.** The baselines were trained without PC-WavKAN’s class-balanced

sampler and augmentation (§3.5), so the parity is between complete methods, and a common imbalance recipe could change the comparison. The B-Spline KAN baseline is a single-bump variant, not a grid-based spline KAN. Latency is x86 CPU only.

Future work. The results point to three specific questions: whether a causal variant (preceding intervals only, a truncated window and a causal filter) preserves the parity; whether the wavelet-KAN family’s behaviour holds on 12-lead recordings with beat-level labels; and whether any behavioural test links individual wavelet parameters to class-specific evidence.

6. Conclusion

This paper introduced WavKAN-CL, a structurally transparent, parameter-efficient arrhythmia classification framework. By replacing fixed activation functions of standard MLPs with learnable Mexican Hat wavelets, the model enables representations that are related to the morphology of the QRS complex, thus solving the black box interpretability challenge [7]. Evaluated on a strict inter-patient protocol (DS1/DS2 split), WavKAN-CL achieves a mean Ventricular (V) recall of 0.88 across We evaluated a 153,045-parameter wavelet Kolmogorov–Arnold network for inter-patient ECG heartbeat classification under the de Chazal protocol, with 20 seeds (peak 0.93), while using only 95,189 parameters. This is a >95% reduction in size when compared to paired seeds, multiplicity correction, effect sizes, confidence intervals and a validation-based architecture selection rule, applied after DS2 had been inspected. Its DS2 Macro-F1 of 0.357 ± 0.019 is not statistically distinguishable from four standard baselines of 30K–118K parameters, and all five models share a poor supraventricular operating point. In the configuration in which components were ablated, structured initialisation of the wavelet parameters helped relative to an isotropic one, the mother wavelet did not measurably matter, and self-attention over interval features was harmful. The parameters do not support a physiological reading once exchangeability and a trained null are considered. Class-balanced sampling raised DS2 supraventricular recall but not on the validation partition. Under preprocessing matched to training, the Transformer-based architectures and supporting its suitability for battery-constrained wearable devices [3]. The discrimination of minority Supraventricular (S) beats remains challenging under non-augmented inter-patient schemes [8], whereas the proposed curriculum learning strategy substantially reduces the generalization gap to ≤ 0.01 . Future work will focus on incorporating P-wave specific attention heads to resolve S-beat ambiguity and deploying the quantized model on low-power edge hardware. These results show that wavelet-based networks that have structural interpretability offer a viable alternative to large black-box models in safety-critical biomedical applications.

Declarations

Funding: Article Processing Charge (APC) of this manuscript was funded by the Vellore Institute of Technology (VIT).

Conflict of Interest: The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper. wavelet KAN transferred worse than both convolutional baselines to INCART and SVDB. We report these null and negative findings alongside the positive ones because, on this evidence, they are the more transferable part of what the study found.

CRedit Author Statement:

CRedit authorship contribution statement

Venkate Ramanan Venkateramanan Manivannan: Conceptualization, Methodology, Software, Validation, Formal Analysis, Investigation, Data Curation, Writing - Original Draft, curation, Writing – original draft, Visualization. **Ramanathan Lakshmanan:** Conceptualization, Resources, Validation, Supervision, Project Administration, Funding Acquisition, Writing - Review & Editing, administration, Funding acquisition, Writing – review & editing.

Data Availability: The datasets analyzed during the current study (MIT-BIH Arrhythmia Database)

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Funding

The Vellore Institute of Technology (VIT) funds the article processing charge. This research received no other specific grant.

Ethics statement

This study uses only publicly available, de-identified PhysioNet databases [28, 29]. No new human-subject data were collected, and no institutional review board approval was required.

Data availability

All datasets are publicly available in the PhysioNet repository: [27] from PhysioNet [28, 29]: the MIT-BIH Arrhythmia Database [27] (<https://physionet.org/content/mitdb/1.0.0/>), the St Petersburg INCART 12-lead Arrhythmia Database (<https://physionet.org/content/incartdb/1.0.0/>), the MIT-BIH Supraventricular Arrhythmia Database [38] (<https://physionet.org/content/svdb/1.0.0/>) and PTB-XL [39] (<https://physionet.org/content/ptb-xl/1.0.3/>).

Code Availability: Source code and pre-trained weights are available at:

Code availability

<https://github.com/vrhsr/WavKAN-CL> contains the source code, preprocessing and training scripts, statistical and figure code, the per-seed test metrics of every arm reported here, the training histories and confusion matrices of every PC-WavKAN arm, the per-seed confusion matrices of all five models from the inference-only re-evaluations, the 20 checkpoints of the evaluated PC-WavKAN configuration, and the 20 checkpoints of its isotropic-initialisation ablation arm used for the prior-retention null. Baseline checkpoints are available from the authors on request. `src/verify_manuscript_numbers.py` re-derives the quantitative claims of this manuscript from those files. The repository name is historical.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the authors used Claude (Anthropic) in order to assist with writing and code development. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

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